Autodesk®
Scaleform®

Scaleform CLIK AS3

Scaleform CLIK

: Prasad Silva Matthew Doyle
: 2.0
: 2010 8 19

Autodesk®
GAMEWARE

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Autodesk® Scaleform® 4.3

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1

Scaleform® Common Lightweight Interface Kit (CLIK™)
CLIK User Guide Getting Started with CLIK Getting Started with CLIK Buttons
2 Scaleform CLIK
CLIK

1.1

Scaleform CLIK ActionScript™ 2.0 (AS) (UI)
Scaleform 4.0
PC UI
(CPU)
UI
CLIK 15 (
)

1.1.1Scaleform CLIK

: UI

Button	Slider
ButtonBar	StatusIndicator
CheckBox	TileList
RadioButton	Label
TextInput	ScrollingList
TextArea	DropdownMenu
ScrollIndicator	NumericStepper
ScrollBar	

: API

InputManager	UIComponent
FocusManager	Tween
DragManager	DataProvider
PopUpManager	IDataProvider
IUIComponent	IList
Constraints	IListItemRenderer

2

Scaleform CLIK UI CLIK

CLIK

CLIK

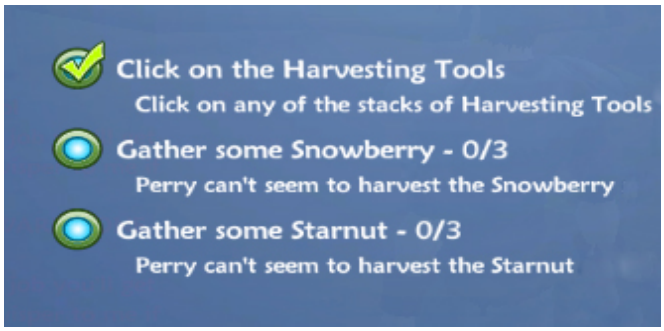
UI

- *:*
- *:* Flash
- *:* (
)
- ***:***
  Flash
- ***:*** UI

2.1

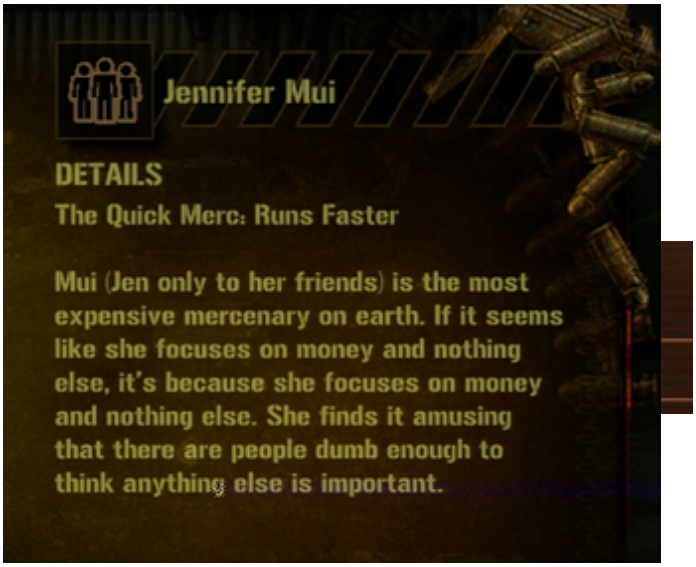
Button CheckBox Label TextInput TextArea
Button CheckBox
Button Label TextInput TextArea
textField

2: *Free Realms –*



3: *Free Realms* -

4: *Free Realms* -



5: *Crysis Warhead* - TextInput

6: *Mercenaries 2* - TextArea

2.1.1 Button

7:

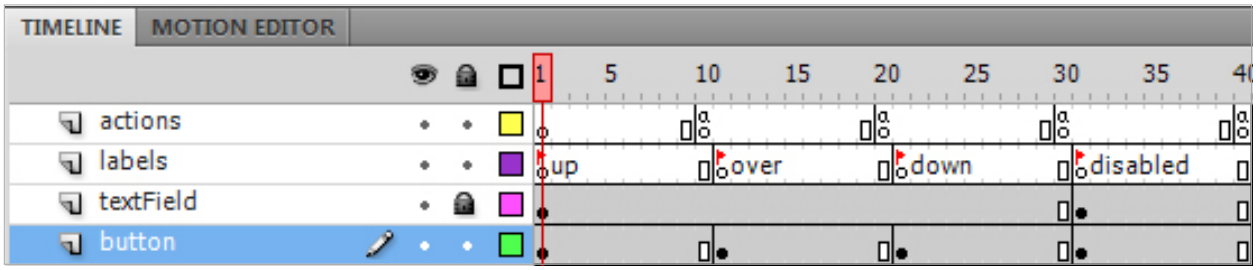
Button CLIK

Button

CLIK Button

:

- *up*
- *over*
- *down*
- *disabled*



9: Button

Flash Button

Getting Started with CLIK Buttons

2.1.1.4

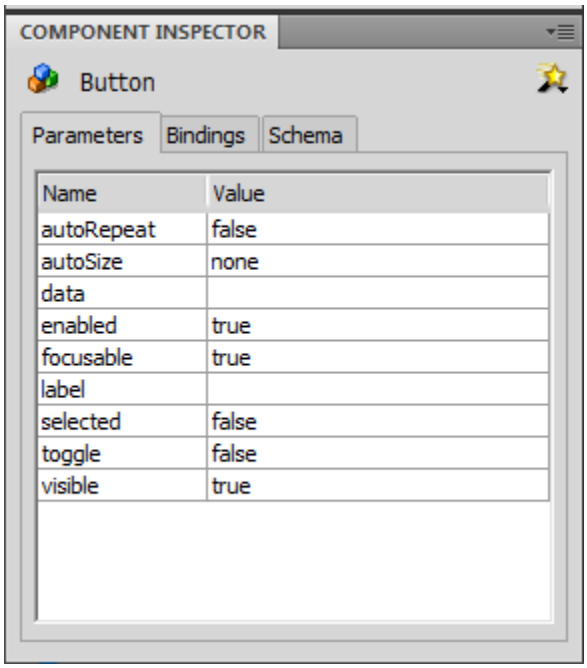


Figure 10: Button component inspectable properties in the CS4 component inspector.

Button :

autoRepeat	
autoSize	autoSize TextFieldAutoSize.NONE
data	ButtonGroup
enabled	false
focusable	/ false Tab
label	
selected	2 toggle true toggle false ActionScript
toggle	Button ( ) true Button
Visible	false

2.1.1.5

1

- ***type***:
- ***target***:
- ***currentTarget***:

- ***eventPhase***: EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***:
- ***cancelable***:

Button

ComponentEvent.SHOW	visible	true
ComponentEvent.HIDE	visible	false
ComponentEvent.STATE_CHA		

	isKeyboard: / true false *isRepeat*: autoRepeating true false
ButtonEvent.DRAG_OVER	( ) *controllerIdx*:
ButtonEvent.DRAG_OUT	( ) *controllerIdx*:
ButtonEvent.RELEASE_OUTSIDE	*controllerIdx*:

ActionScript

```
:myButton.addEventListener( ButtonEvent.PRESS, onButtonPress );
	function onButtonPress( e:ButtonEvent ):void {
		// Do something
	}
```

'myButton' ButtonEvent.PRESS onButtonPress ButtonEvent 'e' Event

2.1.2 CheckBox

11: CheckBox

CheckBox (scaleform.clik.controls.CheckBox)
Button CheckBoxes true/false ( )
ToggleButton toggle

2.1.2.1

CheckBox
/ CheckBox Button

2.1.2.2

CLIK CheckBox MovieClip
:

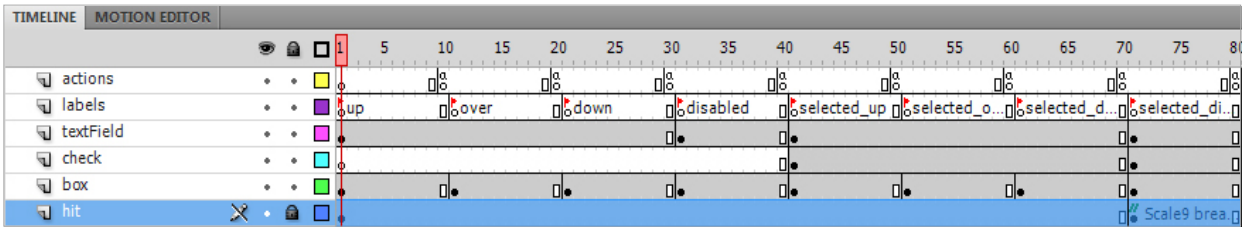
- *textField :* ( ) TextField
- *focusIndicator :* ( ) MovieClip
MovieClip MovieClip "show" "hide" 2

2.1.2.3

toggle CheckBox
:

- *up*
- *over*
- *down*
- *disabled*
- *selected_up*
- *selected_over*

- *selected_down*
- *selected_disabled*

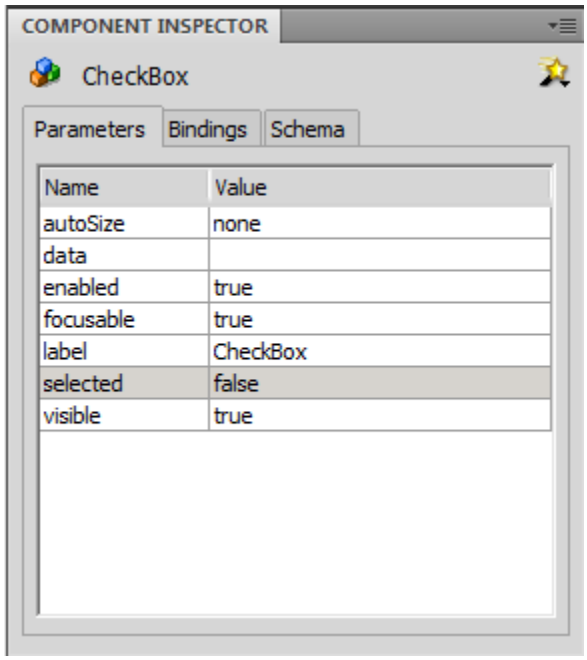


12: CheckBox

CheckBox Button
CheckBox

Getting Started with CLIK Buttons

2.1.2.4



CheckBox Button Button
toggle disableFocus

autoSize	autoSize autoSize=TextFieldAutoSize.NONE
Data	ButtonGroup
Enabled	false
Focusable	/

	false Tab
Label	Button
Selected	2 toggle true toggle false ActionScript
Visible	false

2.1.2.5

1

- ***type***:
- ***target***:
- ***currentTarget***:

- ***eventPhase***: EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***:
- ***cancelable***:

CheckBox

ComponentEvent.SHOW	visible	true
ComponentEvent.HIDE	visible	false
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
Event.SELECT		
ComponentEvent.STATE_CHANGE		
MouseEvent.ROLL_OVER	*mouseIdx*: Scaleform MouseEventEx *buttonIdx*: Scaleform MouseEventEx	
MouseEvent.ROLL_OUT	*mouseIdx*:	

ButtonEvent.RELEASE_OUTSIDE *controllerIdx*:

CheckBox

```
myCheckBox.addEventListener(Event.SELECT, onCheckBoxToggle);
function onCheckBoxToggle(e:Event):void {
        // Do something
}
```

2.1.3Label

13: Label

CLIK Label (scaleform.clik.controls.Label)
MovieClip textField Label
textField

Label textField
Label textField
Label MovieClip textField
textField
Label disabled
textField AS2
Label Label
textField
Label textField
textField
Label

2.1.3.1

Label

2.1.3.2

CLIK Label MovieClip

:

- ***textField*** *:* TextField Label

2.1.3.3

CLIK Label disabled 2 :

- *default*
- *disabled*

14: Label

2.1.3.4

Label :

Text	
visible	false
enabled	false
autoSize	autoSize autoSize=TextFieldAutoSize.NONE

2.1.3.5

1

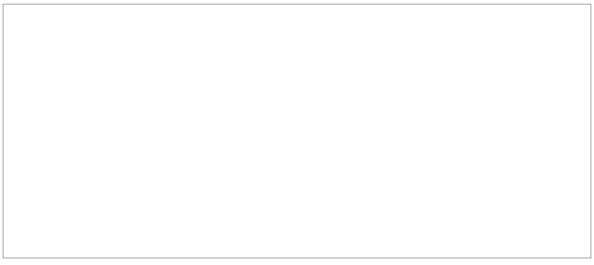
- ***type***:
- ***target***:
- ***currentTarget***:

- ***eventPhase***: EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***:
- ***cancelable***:

Label

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
ComponentEvent.STATE_CHANGE	

2.1.4TextInput



15: TextInput

TextInput (scaleform.clik.controls.TextInput)
textField Label
textField HTML

textField
TextInput textField

TextInput Label
Label TextInput
textField PC UI
TextInput textField

TextInput focused disabled
textField TextInput
textField
textField AS2
TextInput TextInput
Flash Studio
TextInput

TextInput HTML textField

(Shift+) (Shift+Delete) (Ctrl+Insert)
(Shift+Insert)

TextInput
actAsButton inspectable
2
“over” “out” rollOver
rollOut actAsButton “over” “out”
TextInput “default”
CLIK TextInput

Text
String
0xAAAAAA Text
TextFormat

2.1.4.1

TextInput textField

textField

2.1.4.2

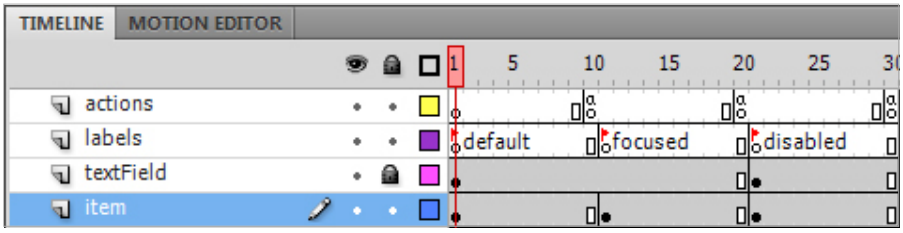
CLIK TextInput MovieClip
:

- ***textField*** *:* textField

2.1.4.3

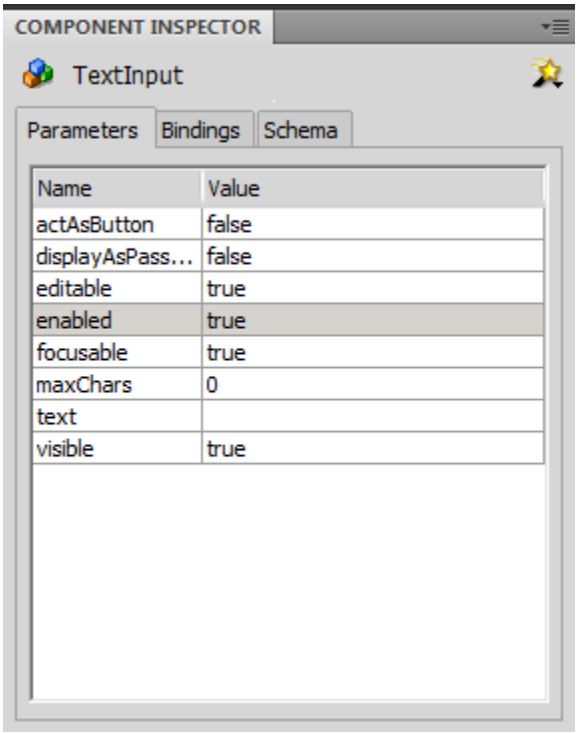
CLIK TextInput focused disabled 3
:

- *default*
- *focused* textField
- *disabled*



16: TextInput

2.1.4.4



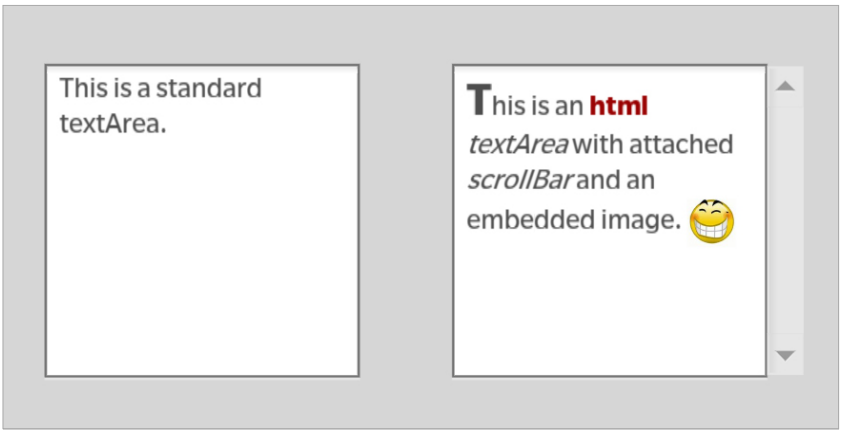
	Scaleform MouseEventEx *buttonIdx*: Scaleform MouseEventEx
MouseEvent.ROLL_OUT	*mouseIdx*: Scaleform MouseEventEx *buttonIdx*: Scaleform MouseEventEx

textField :

```
myTextInput.addEventListener(Event.CHANGE, onTextChange);
function onTextChange(e:Event):void {
        trace("Latest text: " + e.target.text);
}
```

2.1.5TextArea

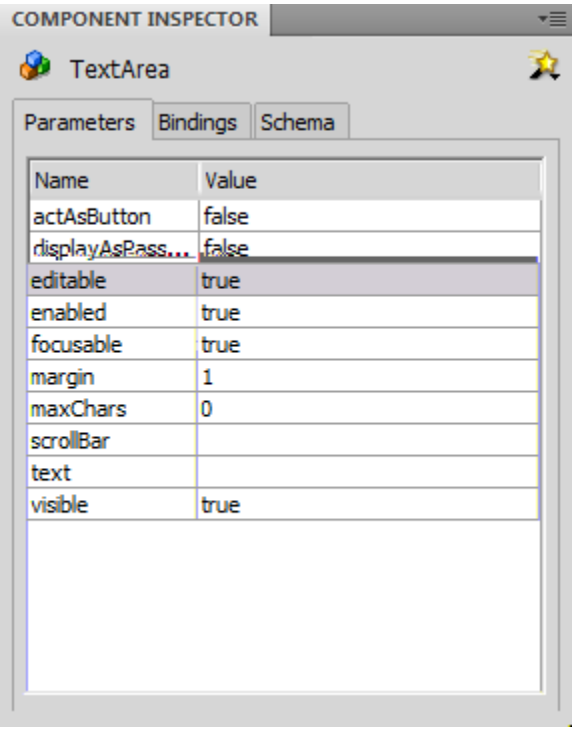
17: TextArea



TextArea (scaleform.clik.controls.TextArea) CLIK TextInput

ScrollBar TextInput TextArea
TextInput

2.1.5.4



TextArea TextInput
password CLIK ScrollBar
2.4 :

actAsButton	true TextInput Button rollOver rollOut Tab TextInput
displayAsPassword	true textField '*' textField
Editable	false TextInput
enabled	false
focusable	/ false Tab
maxChars	textField
scrollBar	CLIK ScrollBar ScrollBar ID (

	TextArea)
scrollPolicy	"auto" "on" ScrollBar "off" ScrollBar (ScrollBar)
text	textField
visible	false

2.1.5.5

1

- ***type***:
- ***target***:
- ***currentTarget***:

- ***eventPhase***: EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***:
- ***cancelable***:

TextArea

ComponentEvent.SHOW		true
ComponentEvent.HIDE		false
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
Event.CHANGE	textField	
Event.SCROLL		
ComponentEvent.STATE_CHANGE		
MouseEvent.ROLL_OVER	*mouseIdx*: Scaleform MouseEventEx *buttonIdx*: Scaleform MouseEventEx	
MouseEvent.ROLL_OUT	*mouseIdx*:	

Scaleform
MouseEventEx
buttonIdx:
Scaleform
MouseEventEx

TextArea :

```
myTextArea.addEventListener(Event.SCROLL, onTextScroll);
function onTextScroll(e:Event):void {
        // Do something
}
```

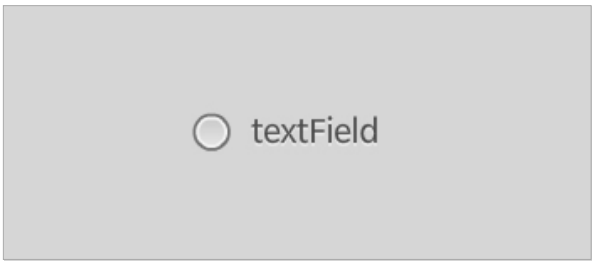
2.2

RadioButton ButtonGroup ButtonBar
ButtonGroup Button

ButtonBar Button RadioButton
ButtonGroup Button

19: *Dawn of War II* -

2.2.1RadioButton



20: RadioButton

RadioButton (scaleform.clik.controls.RadioButton) 1
RadioButton 1
RadioButton

CLIK RadioButton CheckBox
RadioButton
ButtonGroup ( )
RadioButton toggle ( )
ButtonGroup

2.2.1.1

RadioButton
RadioButton
RadioButton ButtonGroup
RadioButton RadioButton Button

2.2.1.2

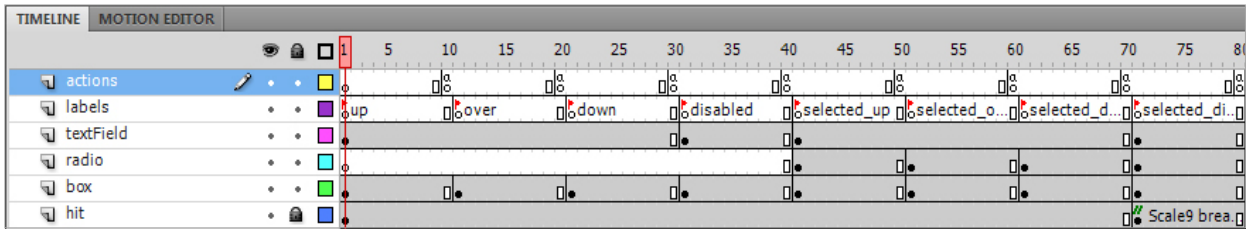
CLIK RadioButton MovieClip
:

- ***textField*** *:* ( ) TextField
- *focusIndicator :* ( ) MovieClip
MovieClip MovieClip "show" "hide" 2

2.2.1.3

RadioButton
CheckBox :

- *up*
- *over*
- *down*
- *disabled*
- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*

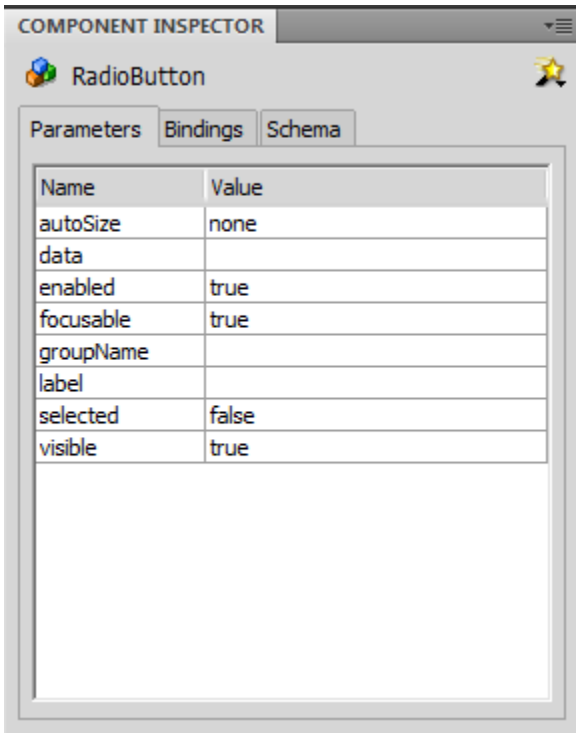


21: RadioButton

RadioButton Button
RadioButton

Getting Started with CLIK Buttons

2.2.1.4



RadioButton Button Button
toggle disableFocus

autoSize	autoSize autoSize=TextFieldAutoSize.NONE
data	ButtonGroup
enabled	false

focusable	/ false Tab
groupName	RadioButton ButtonGroup RadioButton ButtonGroup RadioButton RadioButton in _root ButtonGroup in _root RadioButtons 1
label	
selected	2 toggle true toggle false ActionScript

2.2.1.5

1

- ***type***:
- ***target***:
- ***currentTarget***:

- ***eventPhase***: EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***:
- ***cancelable***:

RadioButton

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	
Event.SELECT	
ComponentEvent.STATE_CHANGE	
MouseEvent.ROLL_OVER	*mouseIdx*:

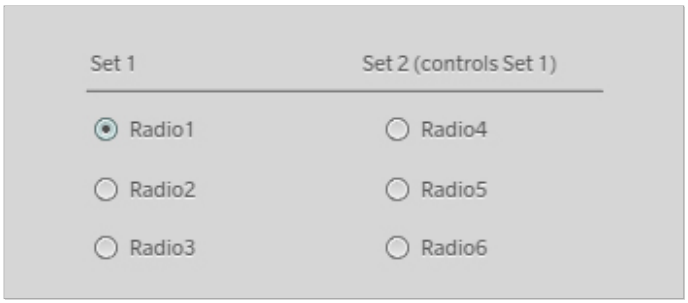
	false *isRepeat*: autoRepeating true false
ButtonEvent.DRAG_OVER	( ) *controllerIdx*:
ButtonEvent.DRAG_OUT	( ) *controllerIdx*:
ButtonEvent.RELEASE_OUTSIDE	*controllerIdx*:

RadioButton :

```
myRadio.addEventListener(Event.SELECT, onRadioToggle);
function onRadioToggle(e:Event):void {
    // Do something
}
```

2.2.2ButtonGroup

22: ButtonGroup



CLIK ButtonGroup (scaleform.clik.controls.ButtonGroup)

ButtonGroup

1

CLIK Button
(CheckBox RadioButton) ButtonGroup

2.2.2.1

ButtonGroup
RadioButton

2.2.2.2

CLIK ButtonGroup MovieClip

2.2.2.3

ButtonGroup

2.2.2.4

ButtonGroup

2.2.2.5

1

- ***type***:
- ***target***:
- ***currentTarget***:

- ***eventPhase***: EventPhase.CAPTURING_PHASE
  EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***:
- ***cancelable***:

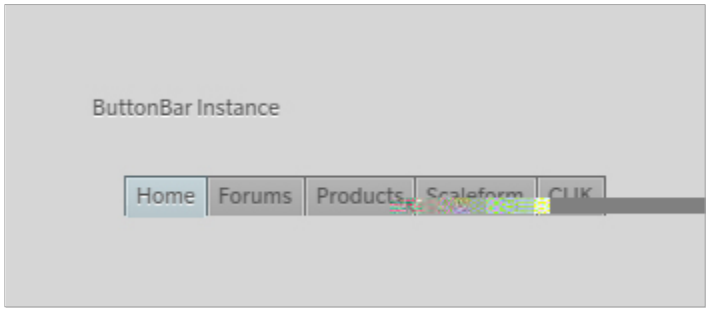
ButtonGroup

Event.CHANGE
ButtonEvent.CLICK
target:

ButtonGroup :

```
myGroup.addEventListener(Event.CHANGE, onNewSelection);
function onNewSelection(e:Event):void {
        if (e.item == myRadio) {
        // Do something
        }
}
```

2.2.3ButtonBar



23: ButtonBar

ButtonBar (scaleform.clik.controls.ButtonBar) ButtonGroup dataProvider Button (dataProvider) ButtonBar UI dataProvider :

```
buttonBar.dataProvider = new DataProvider(["item1", "item2", "item3", "item4", "item5"]);
```

2.2.3.1

ButtonBar ButtonGroup ButtonBar Button

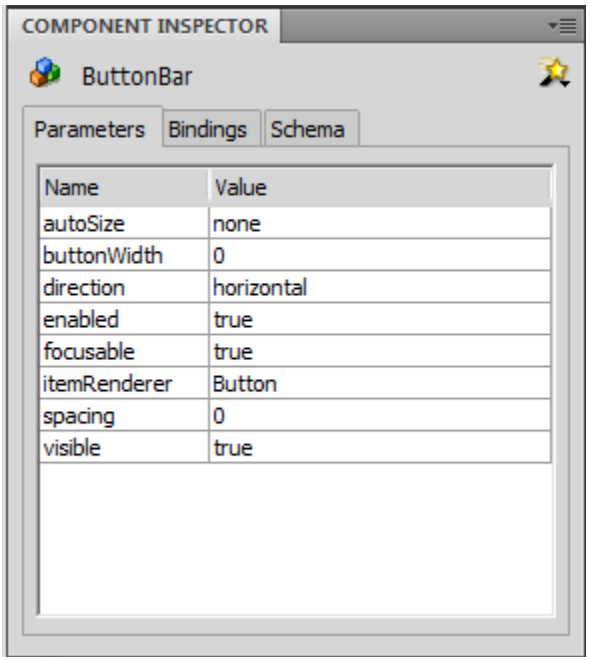
2.2.3.2

CLIK ButtonBar MovieClip

2.2.3.3

CLIK ButtonBar Button

2.2.3.4



ButtonBar (Flash IDE
)
ButtonBar Button

autoSize	true Button
Direction	Button Horizontal Button vertical
buttonWidth	Button autoSize true
enabled	false
focusable	/ false Tab
itemRenderer	Button ID ButtonBar
spacing	Button (*direction*)
visible	false ButtonBar

2.2.3.5

1

- ***type***:
- ***target***:
- ***currentTarget***:

- ***eventPhase***: EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***:
- ***cancelable***:

ButtonBar

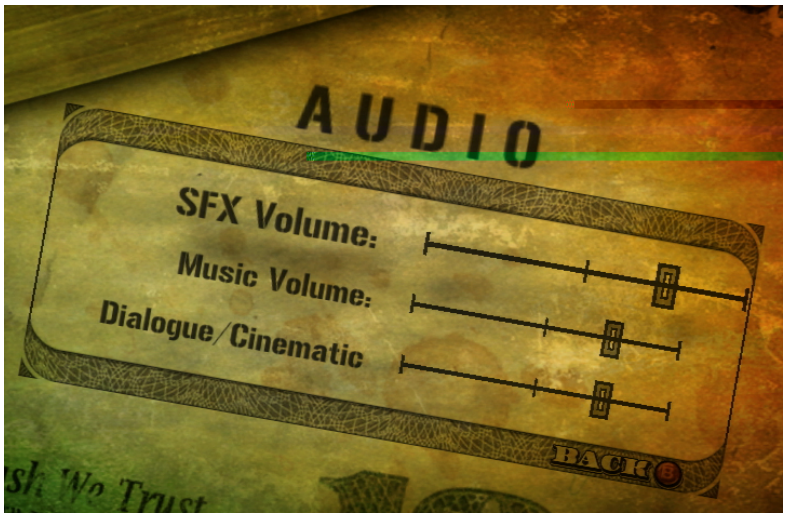
ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	
IndexEvent.INDEX_CHANGE	*index*:ButtonBar. int Values -1 -1 *lastIndex*:ButtonBar. int Values -1 -1 *data*: dataProvider AS3
ButtonBarEvent.BUTTON_SELECT	*index*:ButtonBar Int type Values -1 -1 *renderer*: ButotnBar Button Button

ButtonBar Button
:

```
myBar.addEventListener(ButtonBarEvent.BUTTON_SELECT, onItemClick);
function onItemClick(e:ButtonBarEvent) {
        processData(e.renderer.data);
        // Do something
}
```

2.3

ScrollIndicator ScrollBar Slider
ScrollIndicator
ScrollBar
ScrollBar 4 Button ( )
Slider ScrollBar
Slider
ScrollBar



24: *Mercenaries 2* -

2.3.1ScrollIndicator

25: ScrollIndicator

CLIK ScrollIndicator (scaleform.clik.controls.ScrollIndicator) textField
ScrollIndicator
textField
TextArea ScrollIndicator ScrollBar ( )
ID scrollBar

2.3.1.1

ScrollIndicator

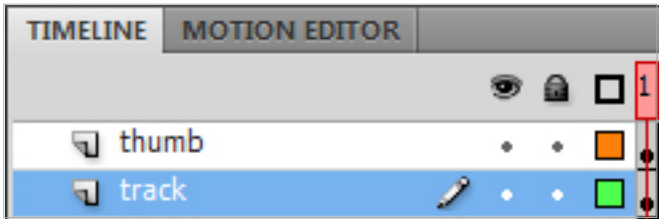
2.3.1.2

CLIK ScrollIndicator MovieClip
:

- *thumb :* CLIK Button
- *track :* MovieClip

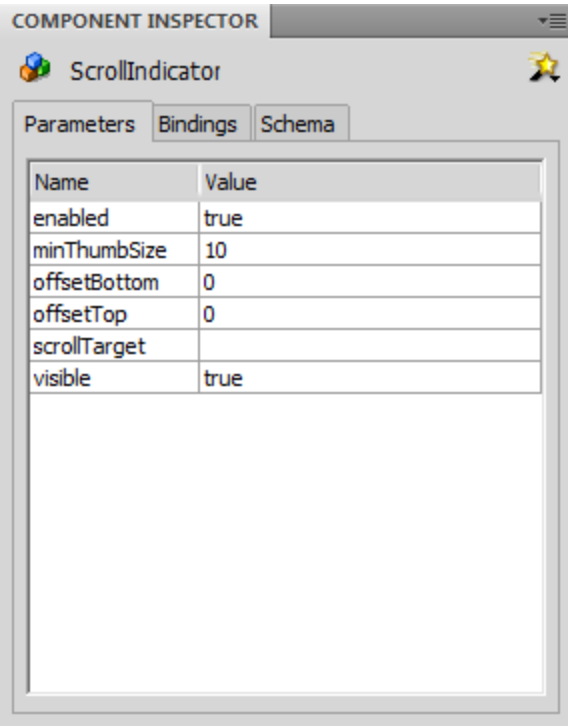
2.3.1.3

ScrollIndicator thumb
track Button



26: ScrollIndicator

2.3.1.4



ScrollIndicator :

Enabled	false	
offsetTop	thumb	
offsetBottom	thumb	
scrollTarget	TextArea	textField textField ScrollIndicator
visible	false	

2.3.1.5

1

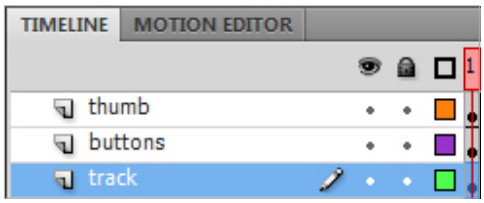
- ***type***:
- ***target***:
- ***currentTarget***:

2.3.2.1

CLIK ScrollBar MovieClip
:

- *upArrow :* CLIK Button Button
- *downArrow :* CLIK Button Button
- *thumb :* CLIK Button
- *track :* CLIK Button

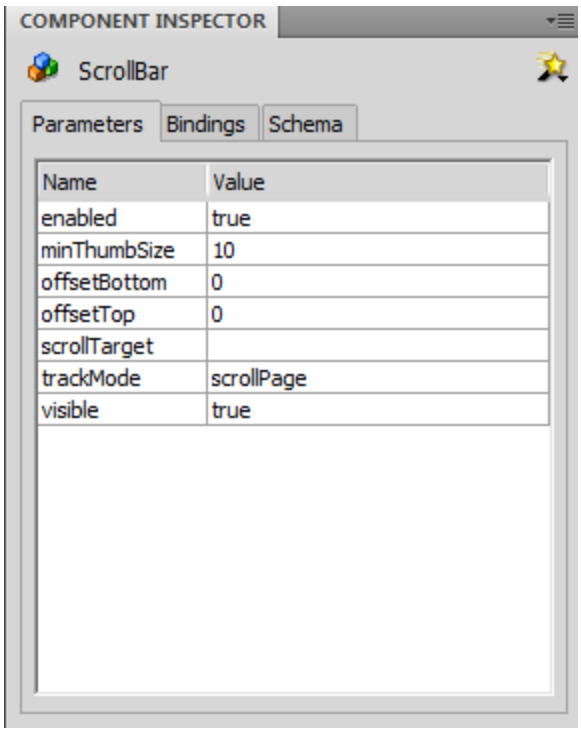
2.3.2.2



28: ScrollBar

ScrollIndicator ScrollBar
thumb up down track Button

2.3.2.3



ScrollBar ScrollIndicator 1 :

enabled	false
offsetTop	thumb
offsetBottom	thumb
scrollTarget	TextArea textField textField ScrollIndicator
trackMode	scrollPage scrollToCursor
visible	false

2.3.2.4

1

- ***type***:
- ***target***:
- ***currentTarget***:

- ***eventPhase***: EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***:
- ***cancelable***:

ScrollBar

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
Event.SCROLL	

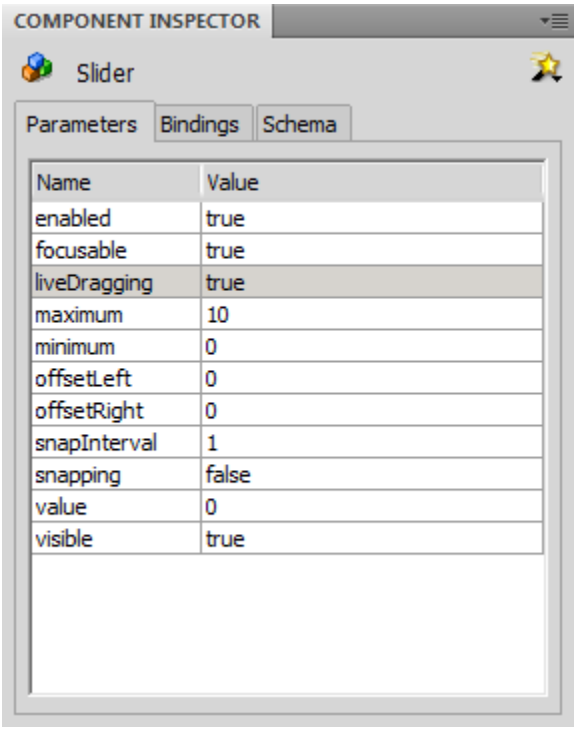
- *thumb :* CLIK Button
- *track :* CLIK Button

2.3.3.3

ScrollIndicator ScrollBar Slider
thumb track Button

30: Slider

2.3.3.4



Slider :

enabled	false		
focusable		false	/ Tab
value	Slider		
minimum	Slider		

maximum	Slider
snapping	true snapInterval
snapInterval	snapInterval snapping false
liveDragging	true Slider change false Slider change
offsetLeft	thumb
offsetRight	thumb
visible	false

2.3.3.5

1

- ***type***:
- ***target***:
- ***currentTarget***:

- ***eventPhase***: EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***:
- ***cancelable***:

Slider

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
ComponentEvent.STATE_CHANGE	
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	
SliderEvent.VALUE_CHANGE	Slider's

Slider :

```
mySlider.addEventListener(SliderEvent.VALUE_CHANGE, onValueChange);
function onValueChange(e:SliderEvent) {
        trace("mySlider.value: " + e.target.value);
        // Do something
```

```
        }
```

2.4

CLIK NumericStepper OptionStepper ScrollingList TileList
DropdownMenu NumericStepper
DataProvider ListItemRenderer
ScrollingList TileList

NumericStepper OptionStepper 1 ScrollingList
TileList 2
ScrollIndicator ScrollBar
DropdownMenu 1
ScrollingList TileList

31: *Mercenaries 2* -

32: *Halo Wars* -

33: *Mass Effect* -

2.4.1 NumericStepper

34: NumericStepper

NumericStepper (scaleform.clik.controls.NumericStepper) 1

2.4.1.1

NumericStepper MovieClip
:

- *textField :* TextField
- *nextBtn :* CLIK Button

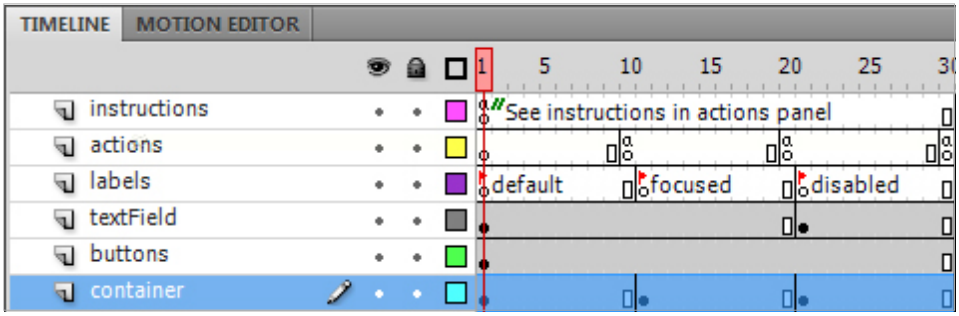
- *prevBtn :* CLIK Button

2.4.1.3

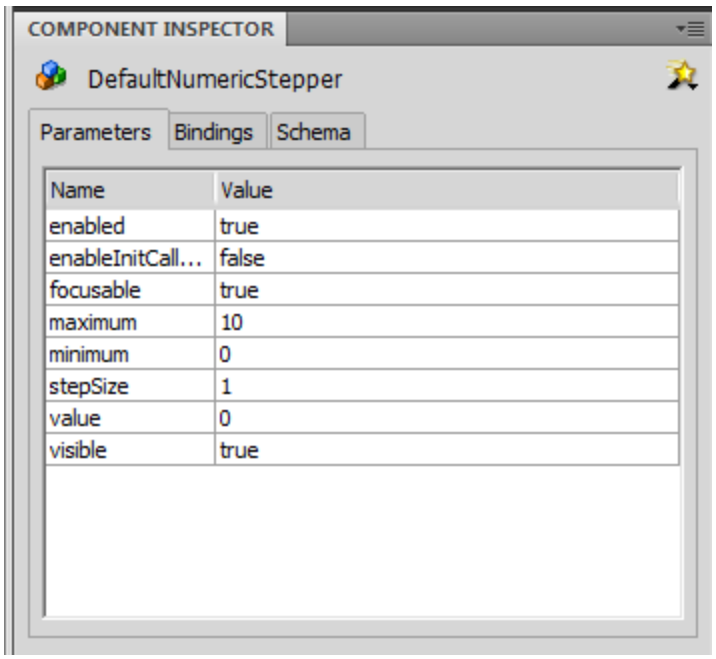
NumericStepper focused disabled 3
:

- *default*
- *focused* textField
- *disabled*

35: NumericStepper



2.4.1.4



NumericStepper MovieClip
:

enabled	false
focusable	/ false Tab
minimum	NumericStepper
maximum	NumericStepper
stepSize	NumericStepper Button 1 NumericStepper /
Value	NumericStepper
visible	false

2.4.1.5

1

- ***type***:
- ***target***:
- ***currentTarget***:

- ***eventPhase***: EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***:
- ***cancelable***:

NumericStepper

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
ComponentEvent.STATE_CHANGE	
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	
ComponentEvent.STATE_CHANGE	
IndexEvent.INDEX_CHANGE	NumericStepper *index*:NumericStepper. int Values -1 -1 *lastIndex*:NumericStepper. int Values -1 -1 *data*: dataProvider AS3

NumericStepper :

```
myNS.addEventListener(IndexEvent.INDEX_CHANGE, onValueChange);
function onValueChange(e:IndexEvent) {
	trace("myNS.value: " + e.target.value);
	// Do something
}
```

2.4.2OptionStepper

36: OptionStepper

NumericStepper OptionStepper (scaleform.clik.controls.OptionStepper) 1
dataProvider

OptionStepper selectedIndex
dataProvider 0-
dataProvider :

```
optionStepper.dataProvider = ["item1", "item2", "item3", "item4"];
```

2.4.2.1

NumericStepper OptionStepper 2

Home End
dataProvider

2.4.2.2

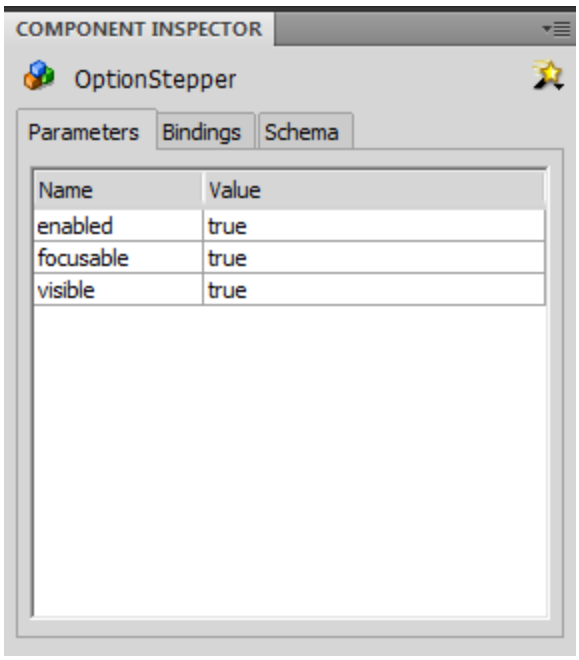
NumericStepper MovieClip
:

- *textField :* TextField
- *nextBtn :* CLIK Button dataProvider

- *prevBtn :* CLIK Button dataProvider

2.4.2.3

OptionStepper



OptionStepper MovieClip
:

enabled	false	
focusable	false	/ Tab
visible	false	

2.4.2.5

1

- ***type***:
- ***target***:
- ***currentTarget***:

- ***eventPhase***: EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***:
- ***cancelable***:

OptionStepper

ComponentEvent.SHOW	true

ComponentEvent.HIDE	false
ComponentEvent.STATE_CHANGE	
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	
IndexEvent.INDEX_CHANGE	OptionStepper *index*:NumericStepper. int Values -1 -1 *lastIndex*:NumericStepper. int Values -1 -1 *data*: dataProvider AS3

OptionStepper :

```
myOS.addEventListener(IndexEvent.INDEX_CHANGE, onValueChange);
function onValueChange(e:IndexEvent) {
    trace("myOS.selectedItem: " + e.target.selectedItem);
    // Do something
}
```

2.4.3ListItemRenderer

38: ListItemRenderer

ListItemRenderer (scaleform.clik.controls.ListItemRenderer) CLIK Button

ScrollingList TileList DropdownMenu

2.4.3.1

ListItemRenderer Button
Button ListItemRenderer

ListItemRenderer

2.4.3.2

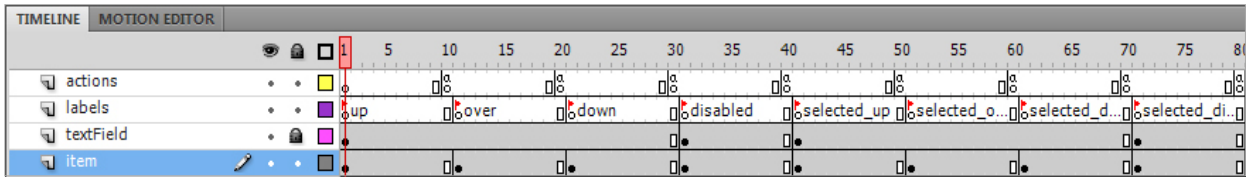
CLIK ListItemRenderer MovieClip
:

- *textField :* ( ) TextField
- *focusIndicator :* ( ) MovieClip
MovieClip MovieClip "show" "hide" 2

2.4.3.3

ListItemRenderer
selected :

- *up*
- *over*
- *down*
- *disabled*
- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*

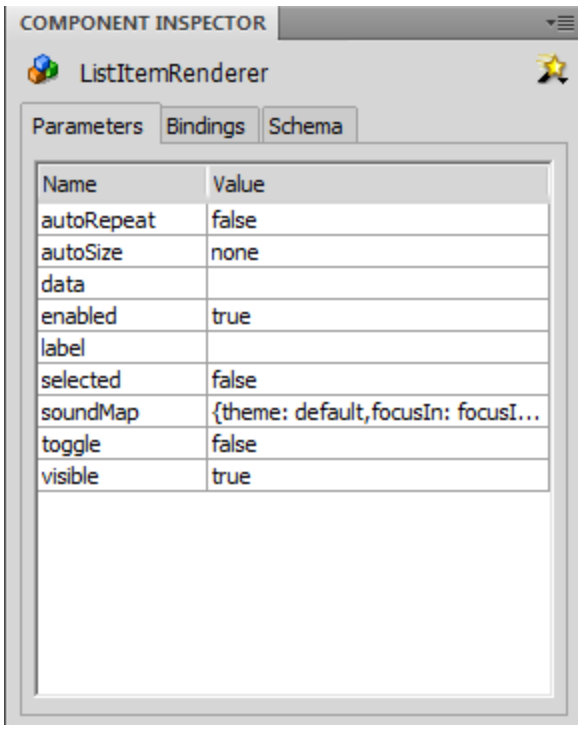


39: ListItemRenderer

ListItemRenderer Button
ListItemRenderer

Getting Started with CLIK Buttons

2.4.3.4



ListItemRenderer
Button

autoRepeat	ListItemRender
autoSize	ListItemRender autoSize autoSize=TextFieldAutoSize.NONE
Data	ListItemRender List component (ScrollingList / TileList) ListItemRender
enabled	false
Label	ListItemRender
selected	ListItemRender ListItemRender 2 ListItemRender toggle true toggle false ActionScript
Toggle	ListItemRender true ListItemRender
Visible	false

2.4.3.5

1

- ***type***:
- ***target***:
- ***currentTarget***:

- ***eventPhase***: EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***:
- ***cancelable***:

ListItemRenderer

ListItemRenderer List List ListItemRenderer ITEM_PRESS ITEM_CLICK ListEvents ListEvent.ITEM_CLICK List ListItemRenderer 1 EventListener EventListener List

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
ComponentEvent.STATE_CHANGE	
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	
Event.SELECT	
MouseEvent.ROLL_OVER	*mouseIdx*: Scaleform MouseEventEx *buttonIdx*: Scaleform MouseEventEx
MouseEvent.ROLL_OUT	*mouseIdx*: Scaleform MouseEventEx *buttonIdx*: Scaleform

	MouseEventEx
ButtonEvent.PRESS	*controllerIdx*: *isKeyboard:* / true false *isRepeat*: autoRepeating true false
MouseEvent.DOUBLE_CLICK	*doubleClickEnabled* true *mouseIdx*: Scaleform MouseEventEx *buttonIdx*: Scaleform MouseEventEx
ButtonEvent.CLICK	*controllerIdx*: *isKeyboard:* / true false *isRepeat*: autoRepeating true false
ButtonEvent.DRAG_OVER	( ) *controllerIdx*:
ButtonEvent.DRAG_OUT	( ) *controllerIdx*:
ButtonEvent.RELEASE_OUTSIDE	

Page Up Page Down

Home End

ScrollBar ScrollingList

ScrollBar ScrollBar

D

CLIK UI

2.4.4.2

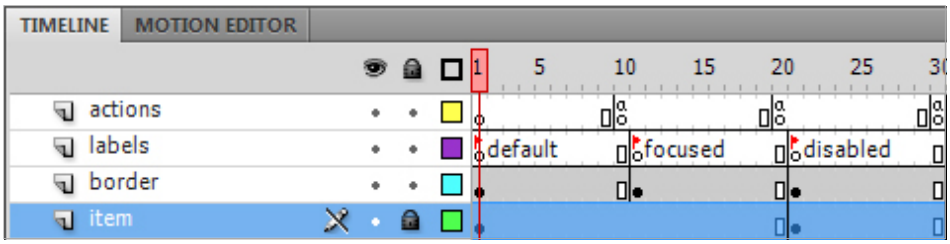
ScrollingList
ScrollingList

2.4.4.3

ScrollingList focused disabled 3

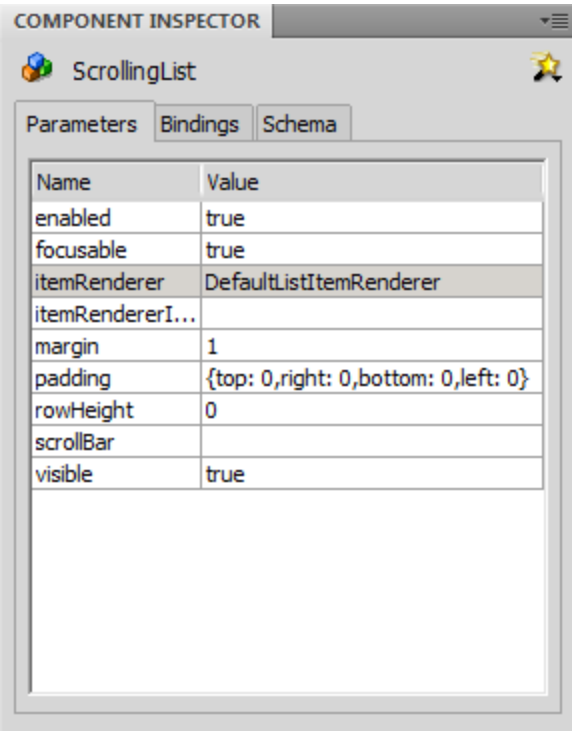
:

- *default*
- *focused*
- *disabled*



41: ScrollingList

2.4.4.4



ScrollingList MovieClip :

enabled	false
focusable	/ false Tab
itemRenderer	ListItemRenderer *itemRendererInstanceName* ListItemRenderers
itemRendererInstanceName	ScrollingList 'r' 'r1' 'r2' 'r3'… 1
margin	*itemRendererInstanceName* ListItemRenderers
padding	

	itemRendererInstanceName ListItemRenderers ScrollBar
rowHeight	*itemRendererInstanceName* ListItemRenderers
Visible	false ScrollBar ListItemRenderers

2.4.4.5

1

- ***type***:
- ***target***:
- ***currentTarget***:

- ***eventPhase***: EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***:
- ***cancelable***:

ScrollingList

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
ComponentEvent.STATE_CHANGE	
FocusHandlerEvent.FOCUS_IN **FocusHandlerEvent.FOCUS_OUT**	
ListEvent.INDEX_CHANGE	• *itemRenderer*: IListItemRenderer • *itemData*: dataProvider ActionScript Object • *index*:list. int Values -1 selectedIndex -1 • *controllerIdx*: /

ListEvent.ITEM_PRESS	• *itemRenderer*: IListItemRenderer • *itemData*: dataProvider ActionScript Object • *index*:list. int Values -1 selectedIndex -1 • *controllerIdx*: /
ListEvent.ITEM_CLICK	• *itemRenderer*: IListItemRenderer • *itemData*: dataProvider ActionScript Object • *index*:list. int Values -1 selectedIndex -1 • *controllerIdx*: /
ListEvent.ITEM_DOUBLE_CLICK	• *itemRenderer*: IListItemRenderer • *itemData*: dataProvider ActionScript Object • *index*:list. int Values -1 selectedIndex -1 • *controllerIdx*: /
ListEvent.ITEM_ROLL_OVER	• *itemRenderer*: IListItemRenderer • *itemData*: dataProvider ActionScript Object • *index*:list. int Values -1 selectedIndex -1 • *controllerIdx*:

ListItemRenderer FLA [ ]

2.4.5.1

ScrollBar TileList

1

TileList 4

TileList

Page Up Page Down

Home End

ScrollBar ScrollingList ScrollBar

ScrollBar

2.4.5.2

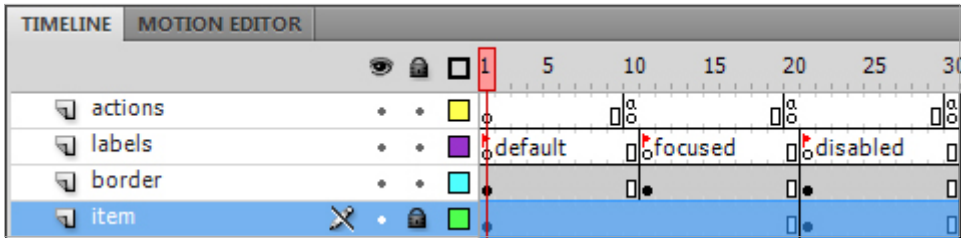
TileList TileList

2.4.5.3

TileList focused disabled 3

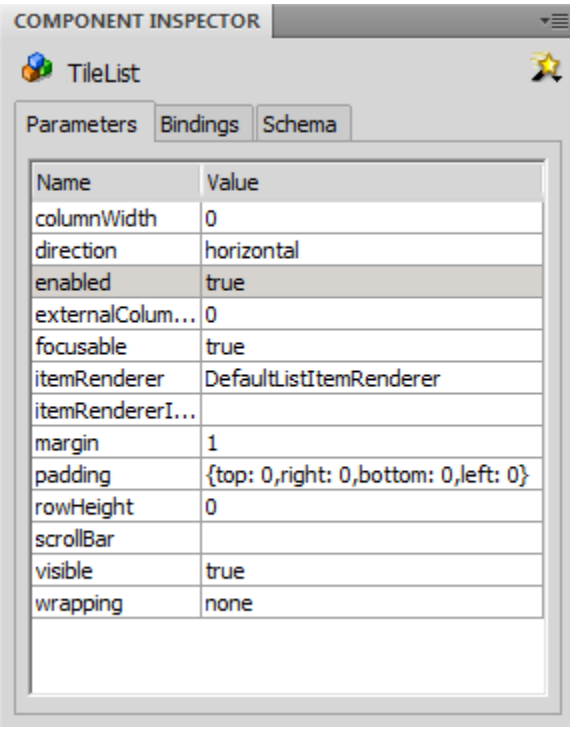
:

- *default*
- *focused*
- *disabled*



43: TileList

2.4.5.4



TileList MovieClip :

columnWidth	*rendererInstanceName*
direction	
enabled	false
externalColumnCount	*rendererInstanceName* TileList
focusable	/ false Tab
itemRenderer	ListItemRenderer itemR*endererInstanceName* ListItemRenderers
itemRendererInstanceName	ScrollingList Stage

	r r1 r2 r3 … 1
margin	*itemRendererInstanceName* ListItemRenderers
padding	itemRendererInstanceName ListItemRenderers ScrollBar
rowHeight	*itemRendererInstanceName* ListItemRenderers
visible	false ScrollBar ListItemRenderers

2.4.5.5

1

- ***type***:
- ***target***:
- ***currentTarget***:

- ***eventPhase***: EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***:
- ***cancelable***:

TileList

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
ComponentEvent.STATE_CHANGE	
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	

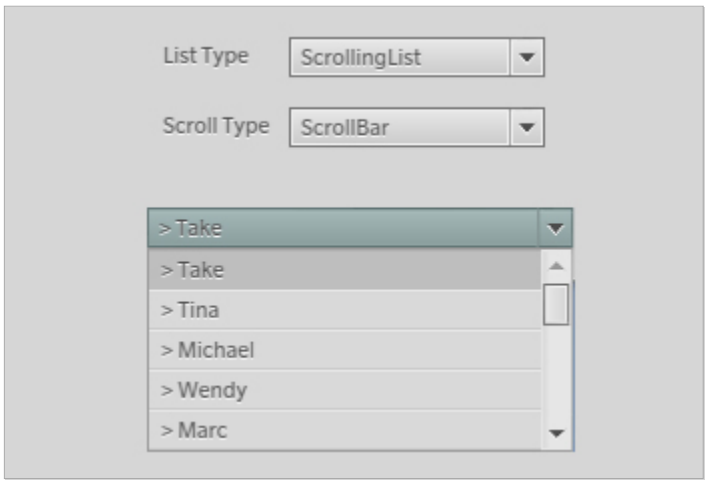
ListEvent.INDEX_CHANGE	• *itemRenderer*: IListItemRenderer • *itemData*: dataProvider ActionScript Object • *index*:list. int Values -1 selectedIndex -1 • *controllerIdx*: /
ListEvent.ITEM_PRESS	• *itemRenderer*: IListItemRenderer • *itemData*: dataProvider ActionScript Object • *index*:list. int Values -1 selectedIndex -1 • *controllerIdx*: /
ListEvent.ITEM_CLICK	• *itemRenderer*: IListItemRenderer • *itemData*: dataProvider ActionScript Object • *index*:list. int Values -1 selectedIndex -1 • *controllerIdx*: /
ListEvent.ITEM_DOUBLE_CLICK	• *itemRenderer*: IListItemRenderer • *itemData*: dataProvider ActionScript Object • *index*:list. int Values -1 selectedIndex -1 • *controllerIdx*:

	/
ListEvent.ITEM_ROLL_OVER	• *itemRenderer*: IListItemRenderer • *itemData*: dataProvider ActionScript Object • *index*:list. int Values -1 selectedIndex -1 • *controllerIdx*: /
ListEvent.ITEM_ROLL_OUT	• *itemRenderer*: IListItemRenderer • *itemData*: dataProvider ActionScript Object • *index*:list. int Values -1 selectedIndex -1 • *controllerIdx*: /

TileList :

```
myList.addEventListener(FocusHandlerEvent.FOCUS_IN, onListFocused);
function onListFocused(e:FocusHandlerEvent) {
    trace("myList is now focused!");
    // Do something
}
```

2.4.6DropdownMenu



44: DropdownMenu

DropdownMenu (scaleform.clik.controls.DropdownMenu)

DropdownMenu
ScrollBar ScrollIndicator ScrollingList TileList
dataProvider
DropdownMenu dataProvider
dataProvider :

```
dropdownMenu.dataProvider = new DataProvider(["item1", "item2", "item3",
"item4"]);
```

DropdownMenu ScrollingList
dropdown
ScrollingList ListItemRenderer FLA [ ]

DropdownMenu
ScrollBar ScrollIndicator DropdownMenu

2.4.6.1

DropdownMenu Enter

ScrollingList TileList ScrollBar
DropdownMenu

DropdownMenu

2.4.6.2

DropdownMenu Button
DropdownMenu MovieClip

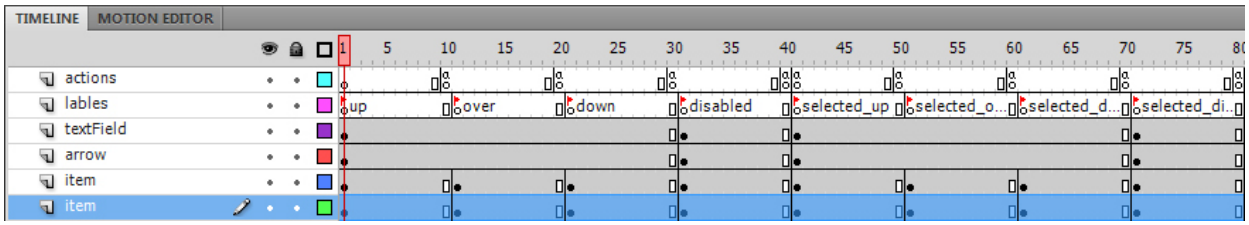
:

- *textField :* ( ) TextField
- *focusIndicator :* ( ) MovieClip
  MovieClip MovieClip "show" "hide" 2

2.4.6.3

DropdownMenu ToggleButton CheckBox
:

- *up*
- *over*
- *down*
- *disabled*
- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*

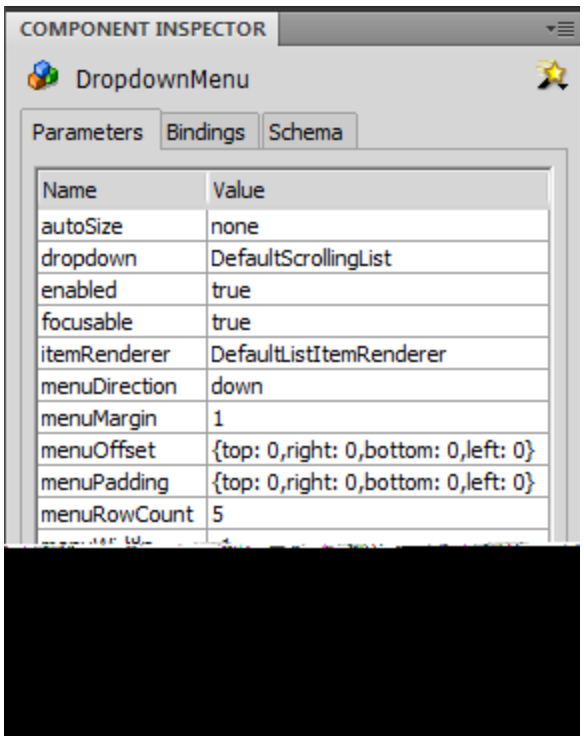


45: DropdownMenu

DropdownMenu Button
DropdownMenu

Getting Started with CLIK Buttons

2.4.6.4



DropdownMenu :

autoSize	DropdownMenu autoSize autoSize="none"
Dropdown	DropdownMenu (ScrollingList TileList)
Enabled	Disables the component if set to false.
Focusable	 false
menuDirection	"up" "down"
menuMargin	
menuOffset	
menuPadding	

menuRowCount	
menuWidth	If set, this number will be enforced as the width of the drop down's list.
thumbOffset	
scrollbar	
Visible	Hides the component if set to false.

2.4.6.5

1

- ***type***:
- ***target***:
- ***currentTarget***:

- ***eventPhase***: EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***:
- ***cancelable***:

DropdownMenu *change* Button

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
ComponentEvent.STATE_CHANGE	
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	
ComponentEvent.SHOW	true
ListEvent.INDEX_CHANGE	• *itemRenderer*: IListItemRenderer • *itemData*: dataProvider ActionScript Object • *index*:list. int Values -1 selectedIndex -1 • *controllerIdx*: /

	isRepeat: autoRepeating true false
ButtonEvent.DRAG_OVER	( ) *controllerIdx*:
ButtonEvent.DRAG_OUT	( ) *controllerIdx*:
ButtonEvent.RELEASE_OUTSIDE	*controllerIdx*:

2.5

( )

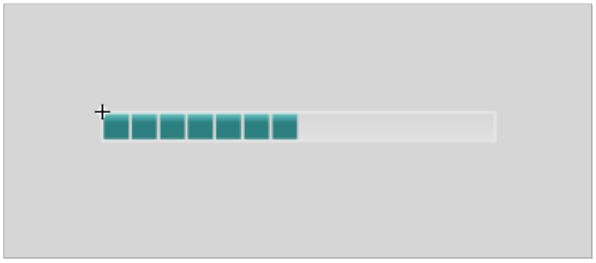
Scaleform CLIK StatusIndicator ProgressBar

2 StatusIndicator

ProgressBar StatusIndicator

46: *Mercenaries 2* - FACTIONS

2.5.1StatusIndicator



47: StatusIndicator

StatusIndicator (scaleform.clik.controls.StatusIndicator)
StatusIndicator

2.5.1.1

StatusIndicator

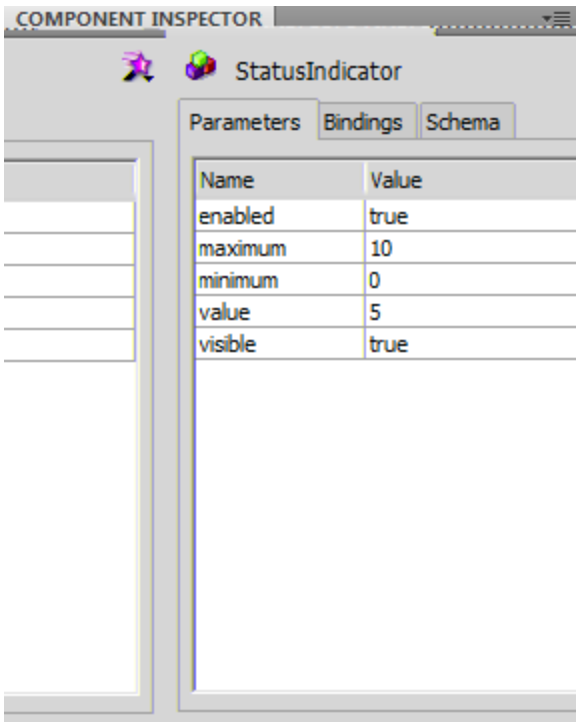
2.5.1.2

CLIK StatusIndicator MovieClip
StatusIndicator 2
stop()
StatusIndicator value
gotoAndStop()

2.5.1.3

StatusIndicator

2.5.1.4



StatusIndicator MovieClip
:

visible	false
enabled	true
value	
minimum	
maximum	

2.5.1.5

1

- ***type***:
- ***target***:
- ***currentTarget***:

- ***eventPhase***: EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***:
- ***cancelable***:

StatusIndicator

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false

2.6

Scaleform CLIK
UI

DragSlot CLIK UI
CLIK
CLIK DragManager scaleform.clik.managers.DragManager.as

CLIK scaleform.clik.controls.Window Alert

PopUpManager PopUpManager.showModal()

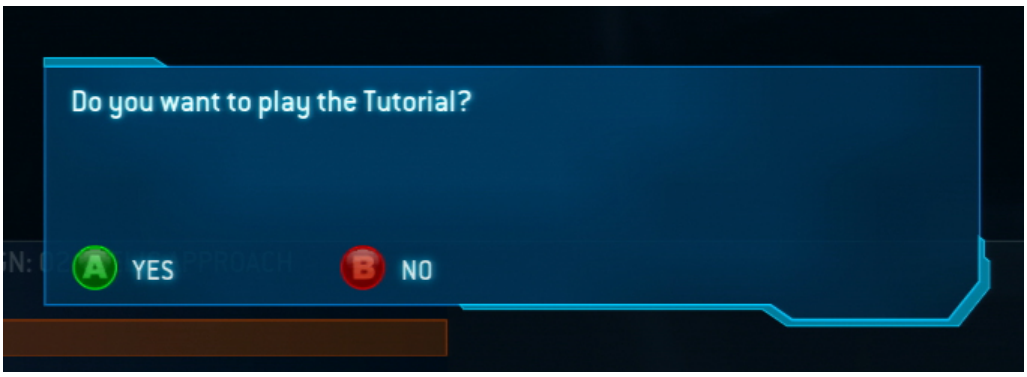


Figure 48: Window example from *Halo Wars*.

2.6.1Window

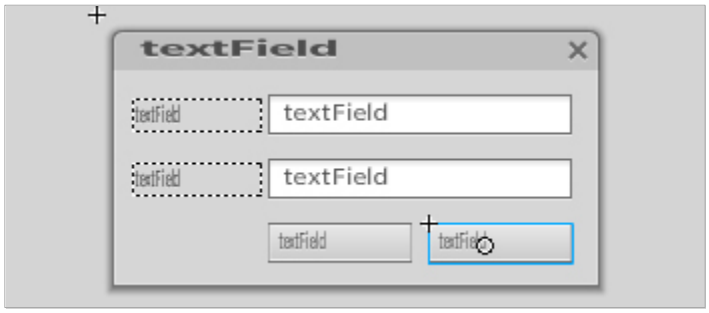


Figure 49: Sample unskinned Window.

CLIK scaleform.clik.controls.Window Alert

MovieClip
MovieClip ( )
1 Dialog.show()

PopUpManager PopUpManager.showModal()

2.6.1.1

Dialog

2.6.1.2

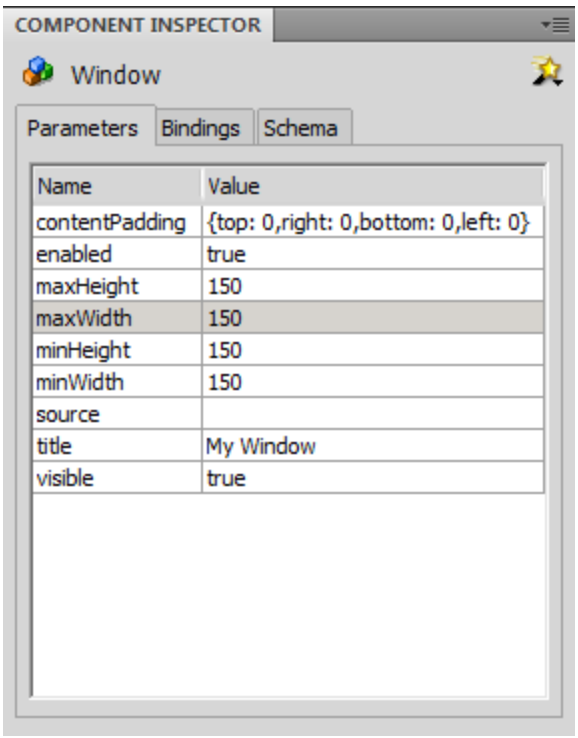
CLIK Dialog MovieClip
:

- ***closeBtn***: CLIK Button Button
- ***titleBtn***: CLIK Button
- ***okBtn***: CLIK Button Accept
  OK
- ***resizeBtn***: CLIK Button
  Button
- ***background*** ( ) MovieClip
- ***hit***:( ) MovieClip hitArea

2.6.1.3

Dialog MovieClip

2.6.1.4



Dialog MovieClip :

contentPadding			
enabled	Disables the component if set to false.		
maxHeight			
maxWidth			
minHeight			
minWidth			
source			
title		titleBtn	String
visible	false		

2.6.1.5

1

- ***type***:
- ***target***:
- ***currentTarget***:

- ***eventPhase***: EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***:

cancelable:
Dialog

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false

:

```
myWindow.addEventListener(ComponentEvent.HIDE, onWindowClosed);
function onWindowClosed(e:ComponentEvent) {
        // Do something.
}
```

3

Scaleform CLIK

3.1

Scaleform CLIK

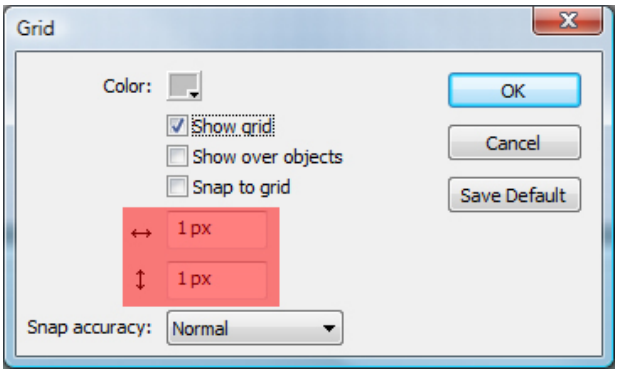
3.1.1

CLIK

Flash

:

1. Flash [ ]
2. [ ] [ ]
3. Flash [ ]
4. [ ] [ ]
5. 1 px
6. [OK]

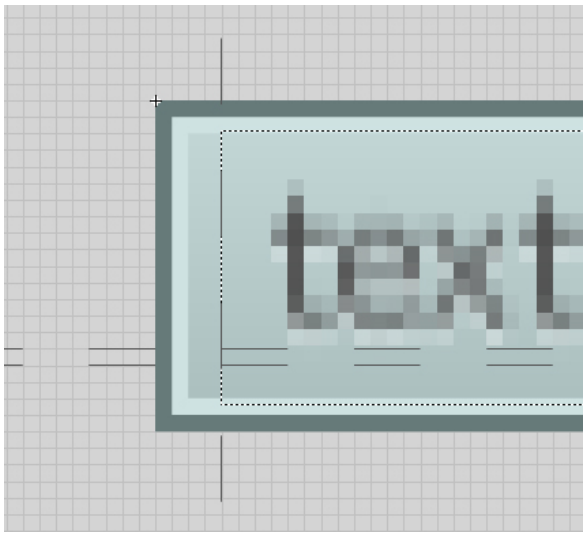


50: [ ]

1

SWF :

2
2



51:

3.1.2

Flash

Scaleform

Flash
PNG Photoshop®

3.1.3

1

( )

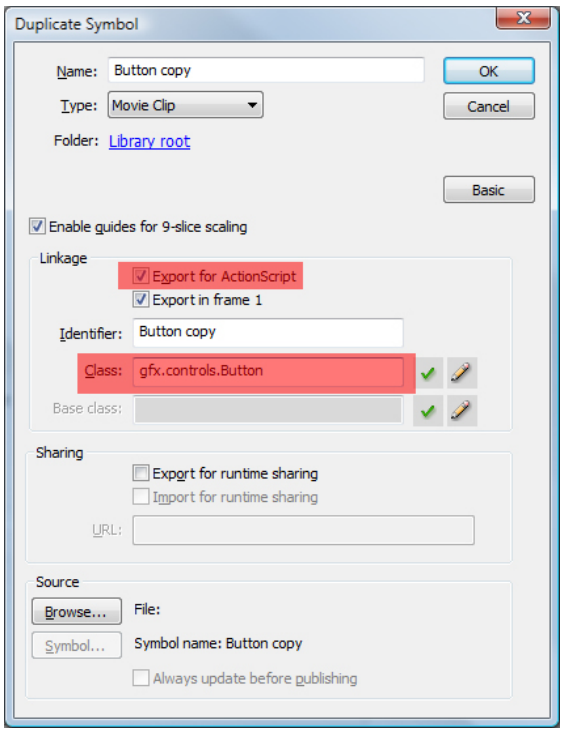
3. FLA [ ] [ ] 'Button'
4. FLA [ ] [ ]
5. 3 [ ]
6. [ ] [ ]

3.2.1.2 (2)

( )

:

1. [ ] [ ]
2. [ ] [ ] (scaleform.clik.controlsButton) Ctrl+C
3. [ ]
4. [ ]
5. [ ] [ActionScript]
6. [ ] Ctrl+V
7. [OK]
8. [ ] [ ]
9. [ ] Ctrl+V
10. [OK]



52: ( )

53: ([ ])

3.3

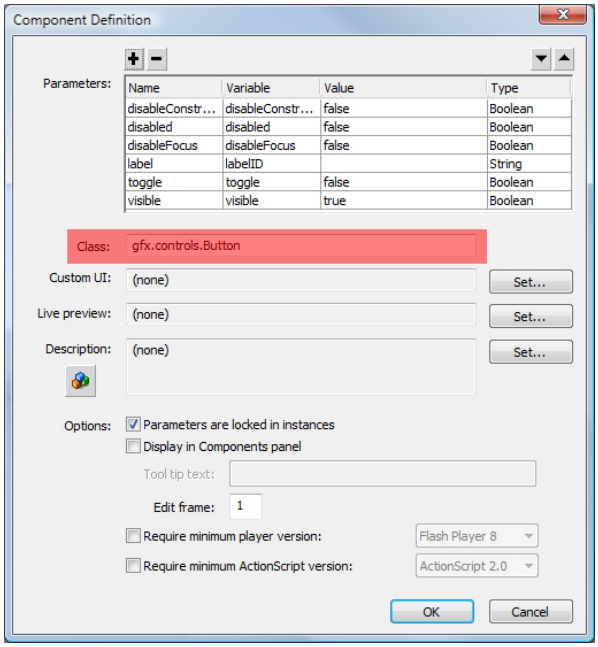
Scaleform CLIK

1

CLIK

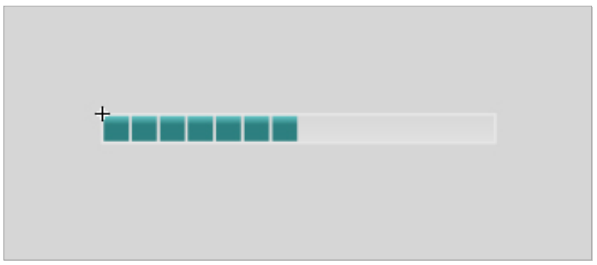
CLIK Flash FLA
[ ]
:

- Flash
- Flash
- Photoshop Illustrator® Flash



Getting Started with CLIK

3.3.1StatusIndicator



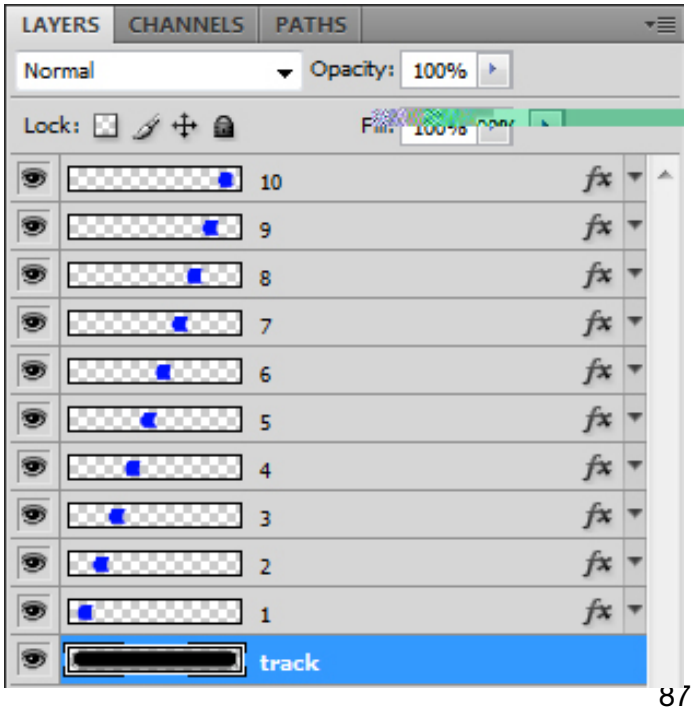
54: StatusIndicator

StatusIndicator Button
StatusIndicator
indicator *track* 2 *track*
indicator

1 Photoshop
StatusIndicator

1
StatusIndicator

55: StatusIndicator PSD



56: StatusIndicator PSD Photoshop

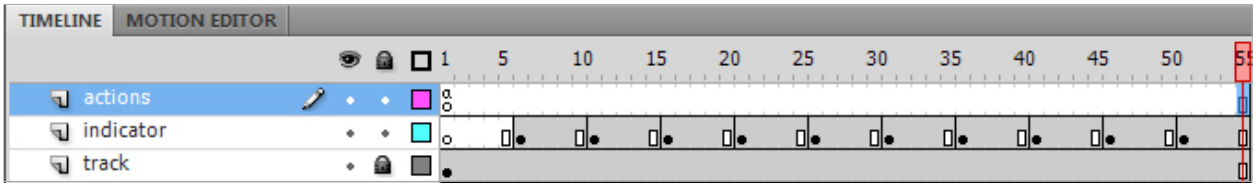
1. StatusIndicator Photoshop

2. PSD
3. Flash [ ] [ ]->[ ]
4. StatusIndicator PSD
5. [ ] [ ] [Flash]

6. [OK]
7. PSD Flash
PSD 10 10
(1) 1 10 1
10 [1]
8.
9.
10.[1] 6
11.[1] 11 11
[ ]
12.[2] Ctrl+X
13.[1] 11 Ctrl+Shift+V
[1]
14.10 (1)
5
[2] 11
[3] 16 [4] 21
10 PSD
51
15. (2–10)
16.
5 [ ]
55

17. *indicator* 。
18. *track* 。 *track*

19.[1] 'indicator'
20. *indicator* *track* *actions* *track*
indicator



57: ( )

21.StatusIndicator
22. 1 10
23.
24.

58: StatusIndicator

3.4

Scaleform CLIK

3.4.1

Scaleform ( ) SWF
Scaleform
Flash UI

3.4.2

Flash Player Scaleform

Scaleform Player Scaleform

Slate Mobile textField Slate Mobile
(CJK) ASCII
Slate Mobile CJK

textField Scaleform (3.4.4
) Scaleform Scaleform

UI

3.4.3textField

textField
( )
textField textField [
] ([ ]->[ ]) [ ]
[OK]
FLA FLA
textField

3.4.4Scaleform

Scaleform
/
SWF SWF
Scaleform

Scaleform textField
textField
gfxfontlib.fla SWF
textField
Scaleform

( ) Scaleform
fontconfig.txt
textField

fontconfig.txt Scaleform
textField
textField '$TITLE'
$TITLE=Scaleform textField
'Scaleform'

Scaleform
Scaleform Font Overview

/ :

- ***enabled;***
- ***visible;***
- ***focused;***
- ***focusable;***
- ***width;***
- ***height;***
- ***scaleX;***
- ***scaleY;***
- *displayFocus :* focused
  true
- ***focusTarget***: Settings this property to another MovieClip will cause that MovieClip to receive focus rather than this component if focus is ever set to this MovieClip. For more information see the Focus Handling section.

UIComponent :

- *configUI :*
- *draw :*

- ***changeFocus :***

- ***scrollWheel** :*

UIComponent Mix-in EventDispatcher

4.1.1

onLoad() :

- MovieClip
- configUI()
- validateNow()
  draw()

4.2

Scaleform CLIK

4.2.1

CLIK
Button ListItemRenderer RadioButton CheckBox ScrollBar

CLIK Button (scaleform.clik.controls.Button)
Button

:

- *up*
- *over*
- *down*
- *disabled*

CLIK Button
"selected_"

selected
:

- *up*
- *over*
- *down*
- *disabled*
- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*

: CLIK
Getting Started with CLIK Buttons

CLIK Button getStatePrefixes()

:

```
protected function getStatePrefixes():Vector.<String> {
    return _selected ? statesSelected : statesDefault;
}
```

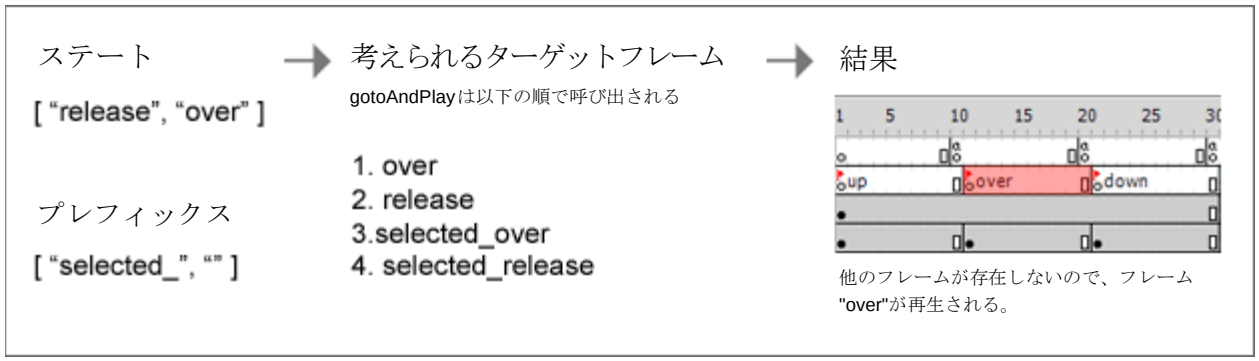
CLIK "selected_"
getStatePrefixes() selected

Button stateMap
CLIK Button
:

```
protected var _stateMap:Object = {
        up:["up"],
        over:["over"],
        down:["down"],
        release: ["release", "over"],
        out:["out","up"],
        disabled:["disabled"],
        selecting: ["selecting", "over"],
        toggle: ["toggle", "up"],
        kb_selecting: ["kb_selecting", "up"],
        kb_release: ["kb_release", "out", "up"],
        kb_down: ["kb_down", "down"]
    }
```

getStatePrefixes()

:



60:

4.2.2

CLIK
ScrollingList OptionStepper Slider TextArea

:

- *default*
- *focused*
- *disabled*

4.2.3

Label

:

- *default*
- *disabled*

4.2.4

StatusIndicator
ProgressBar

0 10 5
(50%) StatusIndicator 50 25 (50
50%) gotoAndStop()
updateValue()

TextInput TextArea
over ***out*** TextInput TextArea
actAsButton

4.3

Scaleform CLIK

ActionScript 3
CLIK

ActionScript 3 Event
http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/events/Event.html

4.3.1

4.3.1.1

addEventListener() removeEventListener()
1

callBack
Function Function

CLIK
Event CLIK
scaleform.clik.events

addEventListener() 3
useCapture **priority** **useWeakReference**

addEventListener()

```
buttonInstance.addEventListener(ButtonEvent.CLICK, onButtonClick, false, 0, true);
```

addEventListener()
http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/events/EventDispatcher.html#addEventListener()

```
buttonInstance.addEventListener(ButtonEvent.CLICK, onButtonClick, false, 0, true);
function onButtonClick (e:Event):void {
        buttonInstance.removeEventListener(ButtonEvent.CLICK, onButtonClick, false);
}
```

buttonInstance ButtonEvent.CLICK
onButtonClick

Event Event
CLIK ( )

4.3.1.2

CLIK dispatchEvent()
1

target

```
dispatchEvent(new Event(Event.CHANGE));
```

4.4

ActionScript 3

```
import flash.utils.getDefinitionByName;
import scaleform.clik.controls.Button;

public var myButton:Button;

// Class
var classRef:Class = getDefinitionByName("ButtonLinkageID") as Class;
//
if (classRef != null) { myButton = new classRef () as Button; }
myButton.addEventListener(ButtonEvent.CLICK, onButtonClick, false, 0, true);
```

FLA Class ButtonLinkageID Base Class
scaleform.clik.controls.Button Symbol

FLA Class Base Class Symbol Symbol
Symbol Class

4.5

Scaleform CLIK

CLIK FocusHandler (scaleform.clik.managers.FocusHandler) 1
FocusHandler

Scaleform
ActionScript ActionScript

- myUIComponent.focused = 1; // Where 1 is a bit for the controller at index 0.
- stage.focus = myMovieClip; // As long as the MovieClip with this logic is on the Stage.
- FocusHandler.getInstance().setFocus(myMovieClip); // FocusHandler is a static class.

4.5.1

InteractiveDisplayObject

(
Tab)
focused true
:

tabEnabled ***mouseEnabled*** MovieClips *false* Scaleform
Flash CLIK
tabEnabled mouseEnabled

tabEnabled mouseEnabled
UIComponent.***focusable***
AS3 MovieClip.focusEnabled
false UIComponent CLIK
focusable tabEnabled mouseEnabled
tabEnabled
mouseEnabled

ScrollBar focusTarget

FocusHandler focusTarget focusTarget

Player
displayFocus
true
Slider Slider
UIComponent.changeFocus()
focused

```
function changeFocus():void {
        track.displayFocus = _focused;
}
```

Tab

tabEnabled
false

```
        background.tabEnabled = false;
```

4.5.2

ScrollingList OptionStepper ButtonBar

Flash Scaleform Selection

:

1. (OptionStepper)
   tabEnabled = mouseEnabled = focusable = false
2. focusTarget
   focusable = true
3. displayFocus
   changeFocus true

4.6

Scaleform CLIK Flash 8 MovieClip
MovieClip

CLIK

4.6.1

InputDelegate
(scaleform.clik.managers.InputDelegate) InputDelegate

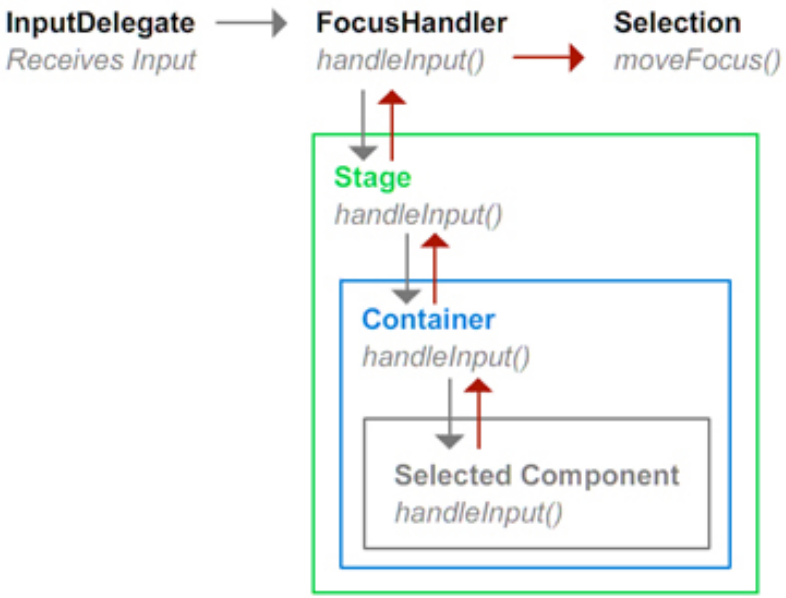
InputDetails (scaleform.clik.ui.InputDetails)
InputDelegate.readInput()
InputDetails InputDelegate

InputDetails :

-
-
-
  key up key down
  InputDetails “keyUp” “keyDown”
-
  navEquivalent ( ) NavigationCode (scaleform.clik.ui.NavigationCode)

FocusHandler InputDelegate InputEvents
InputEvent

handleInput()



61: handleInput()

event.isDefaultPrevented()

```
override public function handleInput(event:InputEvent):void {
        // Check whether the event has already been handled by another component.
        if (event.isDefaultPrevented()) { return; }
        var details:InputDetails = event.details;
```

```
        switch (details.navEquivalent) {
            case NavigationCode.ENTER:
                if (details.value == InputValue.KEY_DOWN) {
                    // Do something.
                    event.handled = true;
                }
                    break;
            default:
                    break;
        }
}
```

InputEvent .bubble ActionScript 3
bubble ScrollingList InputEvent
Enter
ListItemRenderer ButtonEvent.CLICK List

```
override public function handleInput(event:InputEvent):void {
    if (event.isDefaultPrevented()) { return; }
    // Pass on to selected renderer first
        var renderer:IListItemRenderer = getRendererAt(_selectedIndex, _scrollPosition);
    if (renderer != null) {
                // Since we are just passing on the event, it won't bubble, and should properly
stopPropagation.
        renderer.handleInput(event);
         if (event.handled) { return; }
    }
        . . .
}
```

InputDelegate.instance
FocusHandler

```
        InputDelegate.instance.addEventListener(InputEvent.INPUT, handleInput);
        function handleInput(e:InputEvent):void {
                var details:InputDetails = e.details;
                if (details.value = Key.TAB) { doSomething(); }
        }
```

InputDelegate
InputDelegate
W A S D

4.6.2

Wii CLIK
1 SWF

controllerIdx

```
myButton.addEventListener(ButtonEvent.CLICK, onCursorClick);
function onCursorClick(event:ButtonEvent):void {
        trace(event.controllerIdx);
}
```

4.7

62: CLIK

4.7.1

(setter) UIComponent.invalidate()
UIComponent.draw()

draw() isInvalid()

invalidate()
draw()
isInvalid() draw() invalidate()
invalidateData() invalidateSize() invalidate()

invalidate() 1 draw()
Event.RENDER Event.ENTER_FRAME draw()

draw()
UIComponent.validateNow()

4.8

Scaleform CLIK 2 :

1.

2.

scale9grid

Flash 9
Flash Flex “ ”

Scaleform 9
CLIK

CLIK

Flash IDE width height
setSize()
Flash IDE
LivePreview MovieClip

Scaleform Player CLIK
Launcher

4.8.1Scale9Grid

2 Scaleform Player
Scale9Grid MovieClip scale9grid Flash MovieClip
scale9grid Flash Player
Scaleform

1 MovieClip scale9grid
Scale9grid

4.8.2Constraints

Constraints (scaleform.clik.utils.Constraints)

ScrollBar
Constraints

ScrollBar configUI() constraints

draw() constraints
draw()

```
override protected function initialize():void {
        super.initialize();

    // The upArrow doesn't need a constraints, since it sticks to top left.
    if (downArrow) {
        constraints.addElement("downArrow", downArrow, Constraints.BOTTOM);
    }
    constraints.addElement("track", track, Constraints.TOP |
                                    Constraints.BOTTOM);
}

override protected function preInitialize():void {
    constraints = new Constraints(this, ConstrainMode.REFLOW);
}

protected function draw():void {
    constraints.update(_width, _height);
}
```

4.9

dataProvider dataProvider
Scaleform CLIK IDataProvider
(scaleform.clik.interfaces.IDataProvider) API

dataProvider
dataProvider

dataProvider CoreList (ScrollingList TileList) OptionStepper
DropdownMenu

4.9.1

DataProvider (scaleform.clik.data.DataProvider)
initialize() dataProvider ActionScript
dataProvider dataProvider

IDataProvider dataProvider
:

```
myComponent.dataProvider = new DataProvider([ "data1",
                                               4.3,
                                              {label:"anObjectElement", value:6}]);
```

dataProvider :

- *length :* getter
- *requestItemAt :* dataProvider OptionStepper
  1
- *requestItemRange :* dataProvider
  ScrollingList
- *indexOf :*
- *invalidate :* dataProvider
  "dataChange"
  dataProvider
  length
  dataProvider
  ActionScript 2

dataProvider ExternalInterface.call

4.10

Scaleform CLIK Flash 8 Tween Scaleform
Tween (scaleform.clik.motion.Tween) Tween
mx.transitions.easing.*

Scaleform CLIK Tween scaleform.clik.motion.Tween
Tween Scaleform CLIK
Tween fl.transitions.easing.* package

Tween Tween() Sprite
Tween Tween Tween
Object Tween /

- *ease*:
- *easeParam*:Object
- *onComplete*: Tween
- *onChange*: Tween
  /
- *data*:Object Tween Tween

- *nextTween*:Tween Tween
  Tween Tween pause=false
- *frameBased*:

- *delay*:Number .Tween
- *fastTransform*:

- *paused*: Tween

MovieClip

4.10.1

Tween MovieClip MovieClip
tween

Tween Tween 4
Object onComplete
Tween Tween 1

```
import fl.transitions.easing.*;
import scaleform.clik.motion.Tween;

// Perform a 1 second tween from the current horizontal position and
```

Layout Layout_FullScreen_Demo Stage
FxPlayer
Layout_FullScreen_Demo Layout
Layout

4.11.1 **Layout**

Layout
Layout Sprite
Layout Layout
2
myWindow.myMovieClip myWindow.layoutInstance layoutInstance myMovieClip

Layout LayoutData .layoutData Sprite
Layout Layout
.identifier Spitrit LayoutData
.layoutIdentifier

Layout Stage Layout
.layoutData Layout
Stage .layoutData Layout
Sprite/MovieClip Sprite/MovieClip
Stage

public function reset():void
Layout Sprite LayoutData Sprite
Sprite /

resortManagedSprites():void
Sprite
layoutData.layoutIndex Layout
layoutIndex

Flash Symbol Layout
Stage Layout Symbol
0
Layout Layout .rect
Layout

Layout
Sprite/MovieClip LayoutData
4.11.1.1

rect	x y Layout width != 0 Symbol Stage MovieClip x y Layout width == 0 addChild() Symbol x y .rect Rectangle
tiedToStageSize	Layout true false
tiedToParent	Layout true false
hidden	Layout true false Layout visible = false

4.11.2 LayoutData

Sprite scaleform.clik.layout.Layout

4.11.2.1

alignH	Sprite LayoutMode.ALIGN_NONE LayoutMode.ALIGN_LEFT LayoutMode.ALIGN_RIGHT
alignV	Sprite LayoutMode.ALIGN_NONE LayoutMode.TOP LayoutMode.BOTTOM.property TextFieldAutoSize.NONE
offsetH	Layout relativeToH Sprite -1 Layout Stage Layout/Sprite
offsetV	Layout relativeToV Sprite -1 Layout Stage Layout/Sprite
offsetHashH	Layout (Layout.tiedToStageSize == true) "LayoutData.offsetHashH[LayoutData.ASPECT_RATIO_4_3] = 70;" Layout Stage 4:3 Sprite 70px
offsetHashV	Layout (Layout.tiedToStageSize == true)

	"LayoutData.offsetHashV[LayoutData.ASPECT_RATIO_4_3] = 70;" Layout Stage 4:3 Sprite 70px
relativeToH	Sprite Sprite null Sprite
relativeToV	Sprite Sprite null Sprite
layoutIndex	Sprite Layout Sprite relativeToH relativeToV Sprite Layout Sprite layoutIndex (-1)
layoutIdentifier	1 DisplayObjectContainer Layout LayoutData Layout String null Layout Layout

4.12

Scaleform CLIK PopUpManager (scaleform.clik.managers.PopUpManager) Dialog PopUpManager

4.12.1

PopUpManager The createPopUp() MovieClip (CLIK) ID ID String 2 initProperties PopUp Object Object 3 4 x y relativeTo Sprite

Scaleform InteractiveObjectEx.setTopmostLevel()

SWF

topmostLevel

InteractiveObjectEx.setTopmostLevel()
Z

topmostLevel true

4.13

DragManager (scaleform.clik.managers.DragManager)

/

DragEvent.DRAG_STATE
DragManager.handleStartDragEvent()
MouseEvent.MOUSE_UP
DragManager DragManager.handleEndDragEvent()

4.13.1

Figure 63: DragDemo in action.

DragManager UI IDragSlots
IDragSlot DragManager
DragManager

Scaleform CLIK IDragSlot DragSlot
(scaleform.clik.controls.DragSlot)
DragSlot UIComponent CLIK
DragManager DragSlot
MovieClip/Sprite/Bitmap DragSlot Button
DragSlot Button

DragSlot
/ DragSlot
DragSlot dragBegin dragEnd
DragManager DragTarget

DragSlot IDragSlot

DragSlot MMO UI Kit
DragSlot UI

5

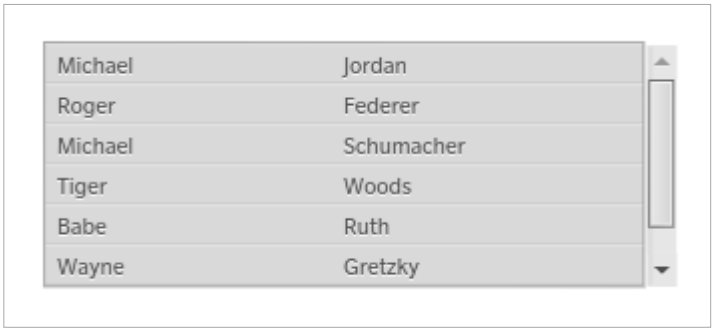
Scaleform CLIK

5.1

Scaleform CLIK

5.1.12 textField ListItem Renderer

64: 2 ScrollingList



ScrollingList ListItemRenderer
ListItemRenderer 1 textField
textField textField
2 textField

2 textField ListItemRenderer
ListItemRenderer ScrollingList
2 textField
dataProvider :

```
list.dataProvider = [{fname: "Michael", lname: "Jordan"},
                     {fname: "Roger",   lname: "Federer"},
                     {fname: "Michael", lname: "Schumacher"},
                     {fname: "Tiger",   lname: "Woods"},
                     {fname: "Babe",    lname: "Ruth"},
                     {fname: "Wayne",   lname: "Gretzky"},
                     {fname: "Usain",   lname: "Bolt"}];
```

dataProvider 2 fname lname
2 2 textField

ListItemRenderer 1 textField 2 textField
ListItemRenderer
MyItemRenderer
ListItemRenderer 2 textField (
FLA *MyItemRenderer.as* :

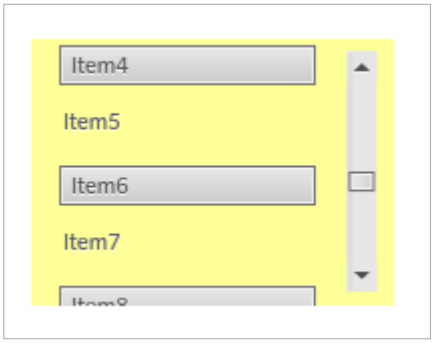
```
import scaleform.clik.controls.ListItemRenderer;

class MyItemRenderer extends ListItemRenderer {

        public var textField1:TextField;
        public var textField2:TextField;

        public function MyItemRenderer() { super(); }

        override public function setData(data:Object):void {
                this.data = data;
                textField1.text = data ? data.fname : "";
                textField2.text = data ? data.lname : "";
        }

        override protected function updateAfterStateChange():void {
                super.updateAfterStateChange();
                textField1.text = data ? data.fname : "";
                textField2.text = data ? data.lname : "";
        }
}
```

ListItemRenderer setData updateAfterStateChange
setData
(ScrollingList TileList)
ListItemRenderer 1 textField MyItemRenderer
textField
updateAfterStateChange
ListItemRenderer
textField

dataProvider
ListItemRenderer
ListItemRenderer
ListItemRenderer 'textField1'
'textField2' 2 textField
MyItemRenderer MyItemRenderer
itemRenderer 'ListItemRenderer'
'MyItemRenderer'

FLA dataProvider fname lname 2

5.1.2



65: CLIK

CLIK ScrollBar

:

- *actions :* ActionScript
- *scrollbar :* CLIK ScrollBar 'sb'
- *mask :* MovieClip
- *content :*
- *background :*

'content' y
0

code ActionScript :

```
// 133 is the view size (height of the mask)
sb.setScrollProperties(1, 0, content.height - 133);
```

```
sb.position = 0;

sb.addEventListener(Event.SCROLL, onScroll, false, 0, true);
function onScroll(e:Event) {
        content.y = -sb.position;
}
```

setScrollProperties

6

1.Flash Studio

```
1152:A conflict exists with inherited definition scaleform.clik... in namespace public.
```

2
FLA "Automatically Declare Stage Instances"
ActionScript FLA

CLIK CLIK
Stage ActionScript

FLA "Automatically Declare Stage Instances" File -> Publish Settings -> "Flash" tab -> Actionscript 3.0 Settings “Automatically declare stage instances” Flash Studio
FLA

2.CLIK .FLA
Stage

Code:

```
1046:Type was not found or was not a compile-time constant:...
```

-
- ActionScript
- ActionScript

ActionScript
myButton:Button (scaleform.clik.controls.)
myButton:MyButtonSymbol MyButtonSymbol
myButton

3. Base Class 2 SWF

Symbol ‘MySymbol’, Layer ‘actions’, Frame 10, Line 1 -- 1024:Overriding a function that is not marked for override.

Symbol ‘MySymbol’, Layer ‘actions’, Frame 20, Line 1 -- 1024:Overriding a function that is not marked for override.

Symbol ‘MySymbol’, Layer ‘actions’, Frame 30, Line 1 -- 1024:Overriding a function that is not marked for override.

Symbol Class Base Class
Class “Button” Symbol Base Class “scaleform.clik.controls.Button”
Symbol Base Class
Base Class Symbol

Symbol Base Class Symbol
Class Base Class
Symbol Class

7

1. Symbol Class Symbol Base Class
   Base Class Symbol Base Class Class
   Base Class Base Class

   Base Class Class
   Symbol Base Class
   Base Class Symbol Base Class
   Symbol getDefinitionByName().

8 CLIK AS3 vs. CLIK AS2

This section outlines major differences, mostly programming related, between CLIK AS2 and CLIK AS3.

1. CLIK AS3 is designed to work in Flash Player.
2. Rather than using a custom EventDispatcher, CLIK AS3 uses ActionScript 3's native event system.
3. The invalidation system has been redesigned to help avoid unnecessary updates in draw(). Each component now stores an "invalid" table that tracks which aspects of the component are currently invalid. Default invalidation types are defined in scaleform.clik.constants.InvalidationType. Aspects can be marked as invalid using invalidate(), invalidate[InvalidationType](), or manually setting the property to true within the table.. Logic within draw() should be wrapped within an appropriate isInvalid() check to ensure that unnecessary updates to components are avoided.
4. invalidate() no longer uses a 1ms delay; instead, it uses the next stage invalidatation (Event.RENDER) or the next Event.ENTER_FRAME for the component and then calls validateNow().
5. Tween's syntax is no longer MovieClip.tweenTo; instead, var t:Tween = new Tween();
6. handleInput() is now tied to the native event system. This means that handleInput() now accepts one parameter of type InputEvent and the event will bubble up from the component that dispatched it rather than being passed down from FocusHandler.
7. handleInput() no longer returns a Boolean; instead, it should set event.handled = true; if the event was handled.
8. UIComponent has a new property, focusable, which can be used with the CLIK FocusManager to prevent focus from reaching a component. This is more or less a replacement for AS2's MovieClip.focusEnabled property.
9. enableInitCallback is no longer an inspectable of UIComponent. Instead, it should be set within a class's constructor or preInitialize() method.
10. DataProvider now requires that you provide the target array to DataProvider's constructor (myComponent.dataProvider = new DataProvider(myArray);), rather than simply setting myComponent.dataProvider = myArray;

11. ButtonBar will now only create Buttons that fit within its own size. If your ButtonBar's size is 200px wide and each Button is 150px wide, only one Button will be displayed. This allows users to resize their ButtonBar and have the Buttons within be added and removed dynamically.
12. UILoader has been removed. This functionality is now handled by the Flash MovieClip/Sprite and Loader classes.
13. UIComponent.SoundMap, introduced in CLIK 3.3, has been removed for CLIK AS3.